

Practice Problem Set 2.2

CSE251 - Electronic Devices and Circuits

CLOSE LOOP CONFIGURATIONS OF OP-AMP

Inv and Non-inv Amplifiers, Inv Adder, Integrator and
Differentiator, Function Implementation, and VTC

For each problem, assume the op-amps to be ideal having infinite open loop gain, zero input currents, and zero output resistance.

[Course Description, COs,
and Policies](#)



[Midterm and Final
Questions](#)

Problem 1

- Design circuits using op-amps to implement the following operational functions. Here, v_1 , v_2 , and v_3 are the inputs and v_o is the output.

I. $v_o = -4v_1 + 5v_2$

II. $v_o = -7v_1 + \frac{d}{dt}v_2$

III. $v_o = \frac{3}{4}v_1 + 7v_2 - \frac{d}{dt}v_3$

IV. $v_o = -7 \int v_1 dt + \frac{4}{3}v_2 - 3 \frac{d}{dt}v_3$

V. $v_o = \int (6v_1 - v_2) dt + 3 \frac{d^2}{dt^2}(v_3)$

VI. $v_o = \int (6v_1 - 3 \frac{d^2v_2}{dt^2}) dt - 3 \frac{dv_3}{dt}$

VII. $v_o = \frac{d}{dt} \left(3v_1 - \frac{3}{2} \int v_2 dt \right) + \int 4v_3 dt$

VIII. ** $v_o = -\frac{1}{3} \int v_1 dt + 2 \ln(v_2) + 4v_3$

IX. ** $f = -3 \frac{d}{dt}v_1 + 2e^{v_2} + 4v_3$

X. ** $f = v_1v_2/v_3$

Note: Problems marked with an asterisk (**) are a bit more advanced for this course. However, attempting them can help you develop a stronger grasp of the topic.

Problem 2

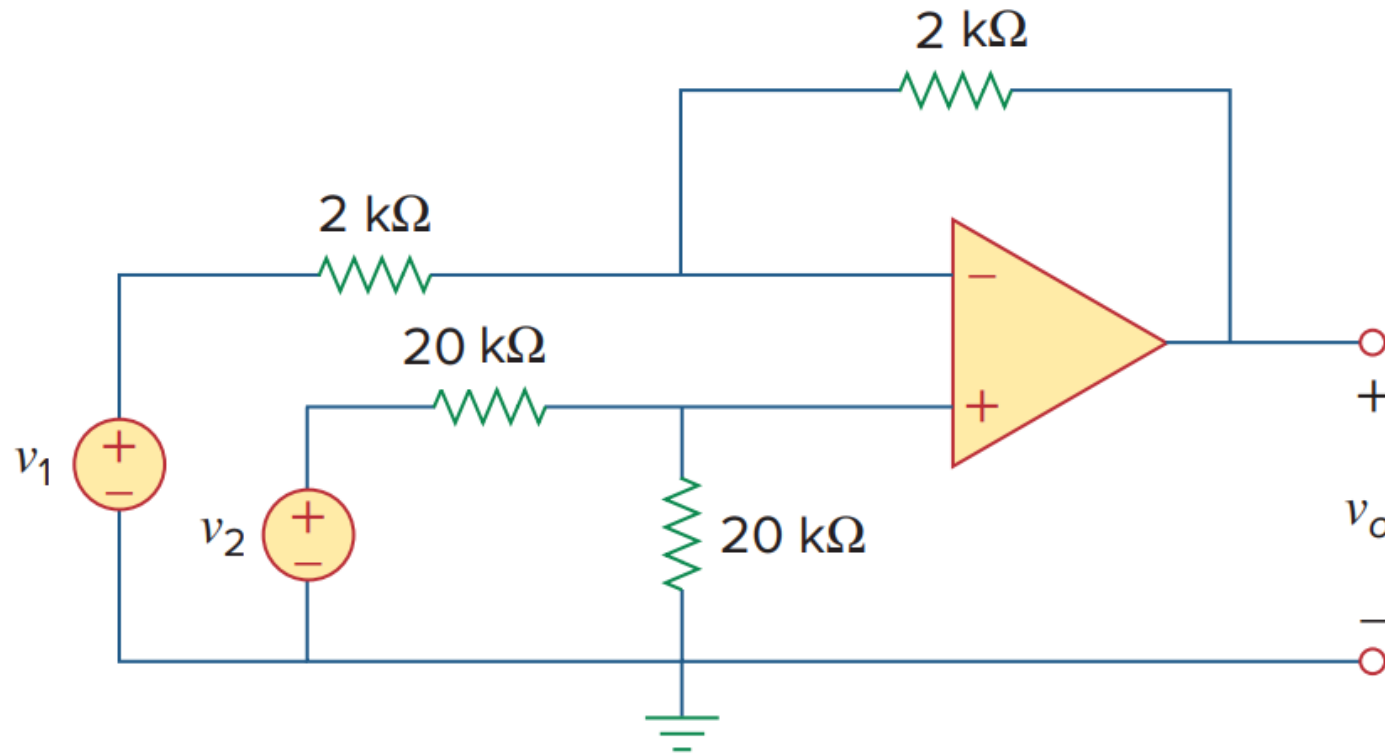
- I. Using not more than two op-amps, design a circuit to implement the following expression. v_1 , v_2 , and v_3 are the inputs and v_o is the output.

$$-v_o = \frac{v_3 - v_1}{5} + \frac{v_1 - v_2}{2}$$

- II. Design a circuit that will average three voltages.

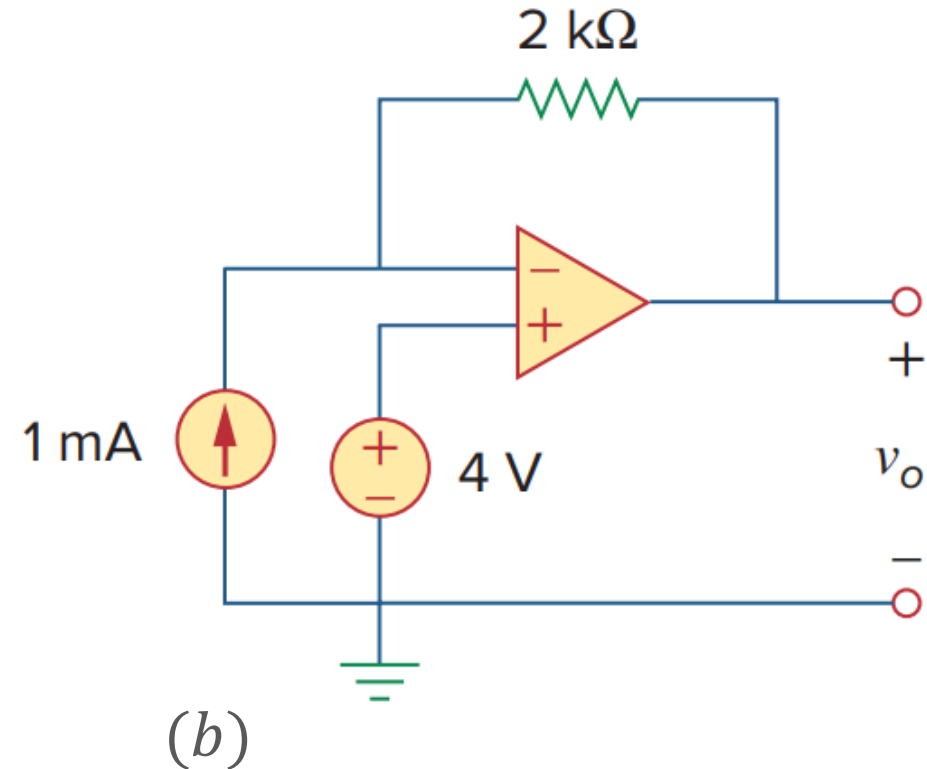
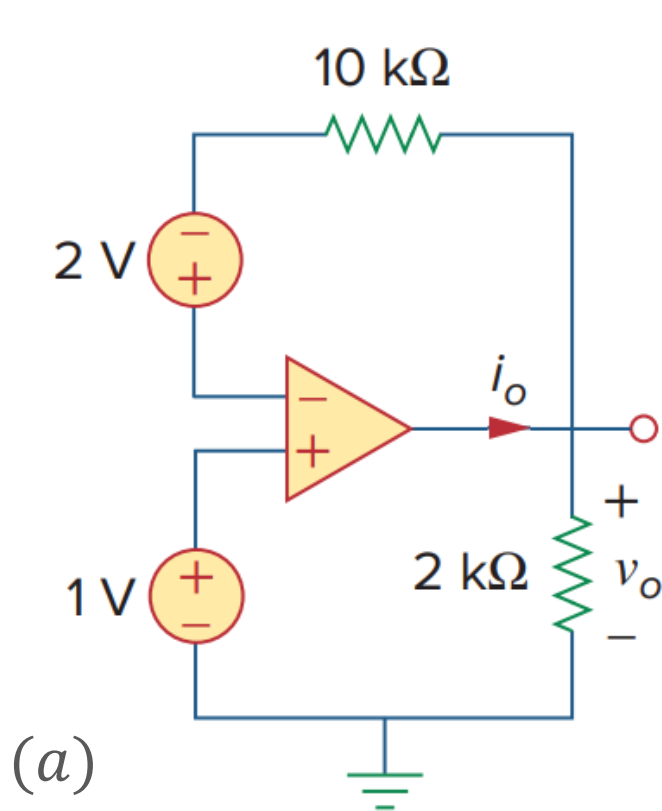
Problem 3

- Prove that the following circuit is a subtractor which produces an output equal to the difference between v_2 and v_1 .



Problem 4

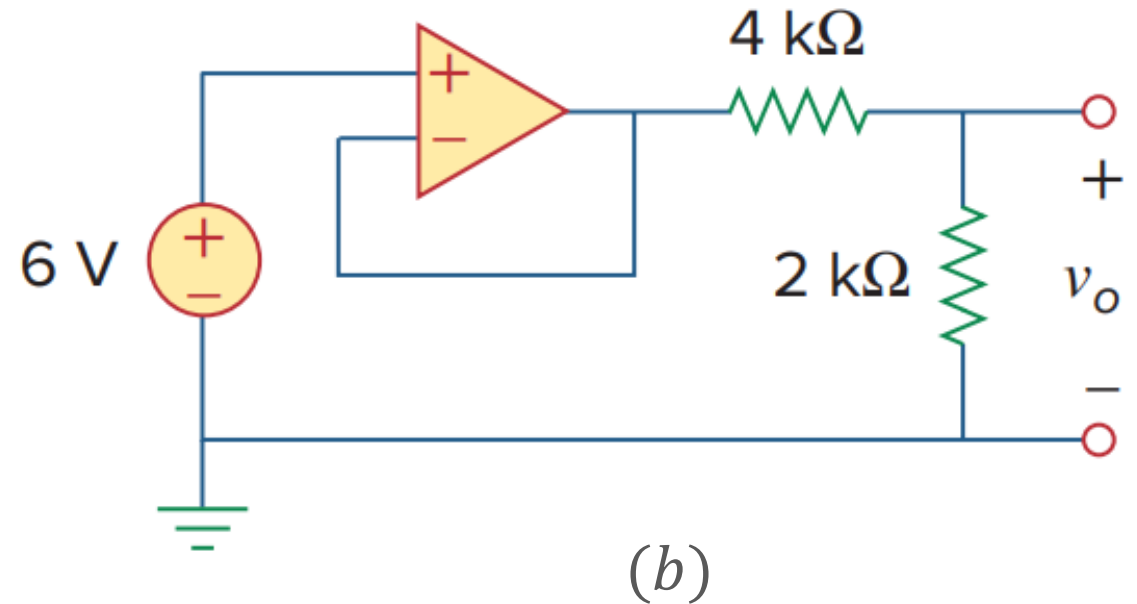
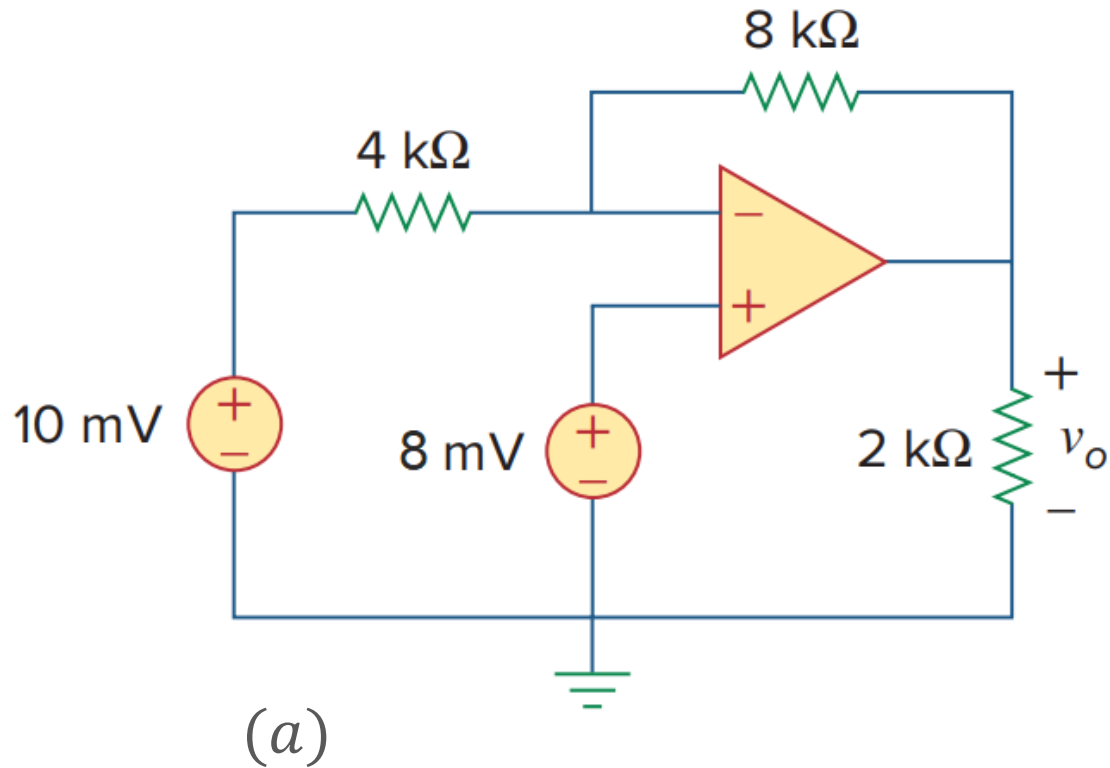
- Determine v_o and i_o for the circuits shown below.



Ans: (a) $v_o = -1\text{ V}$, $i_o = -0.5\text{ mA}$; (b) $v_o = 2\text{ V}$

Problem 5

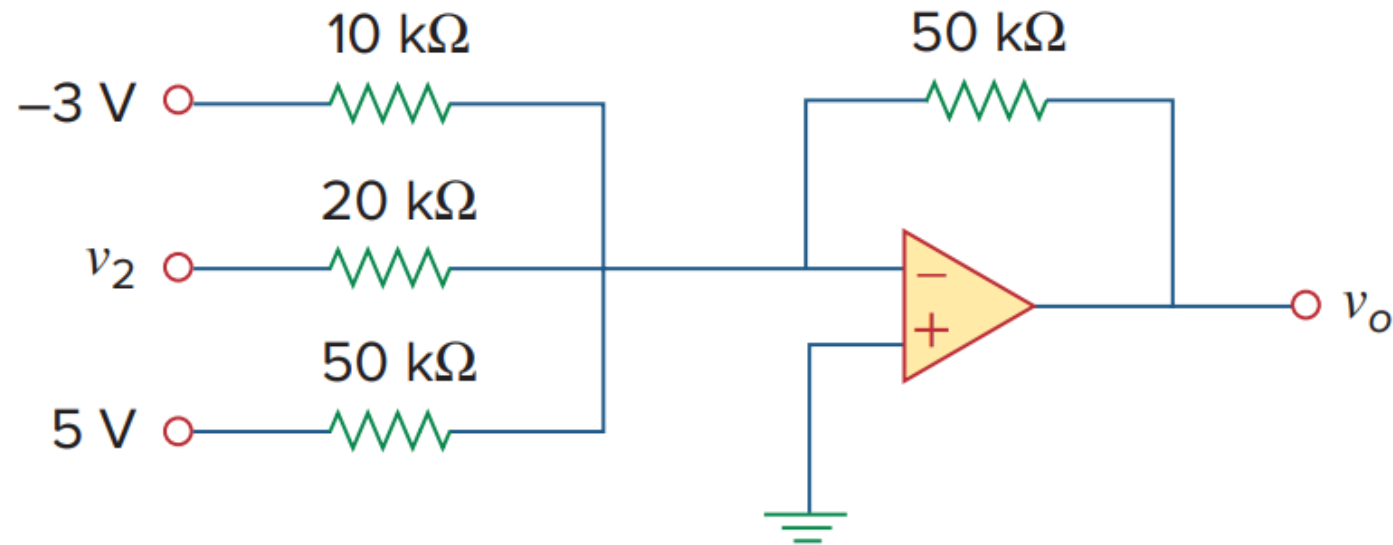
- For the circuits shown below, determine v_o .



Ans: (a) 4 mV; (b) 2 V

Problem 6

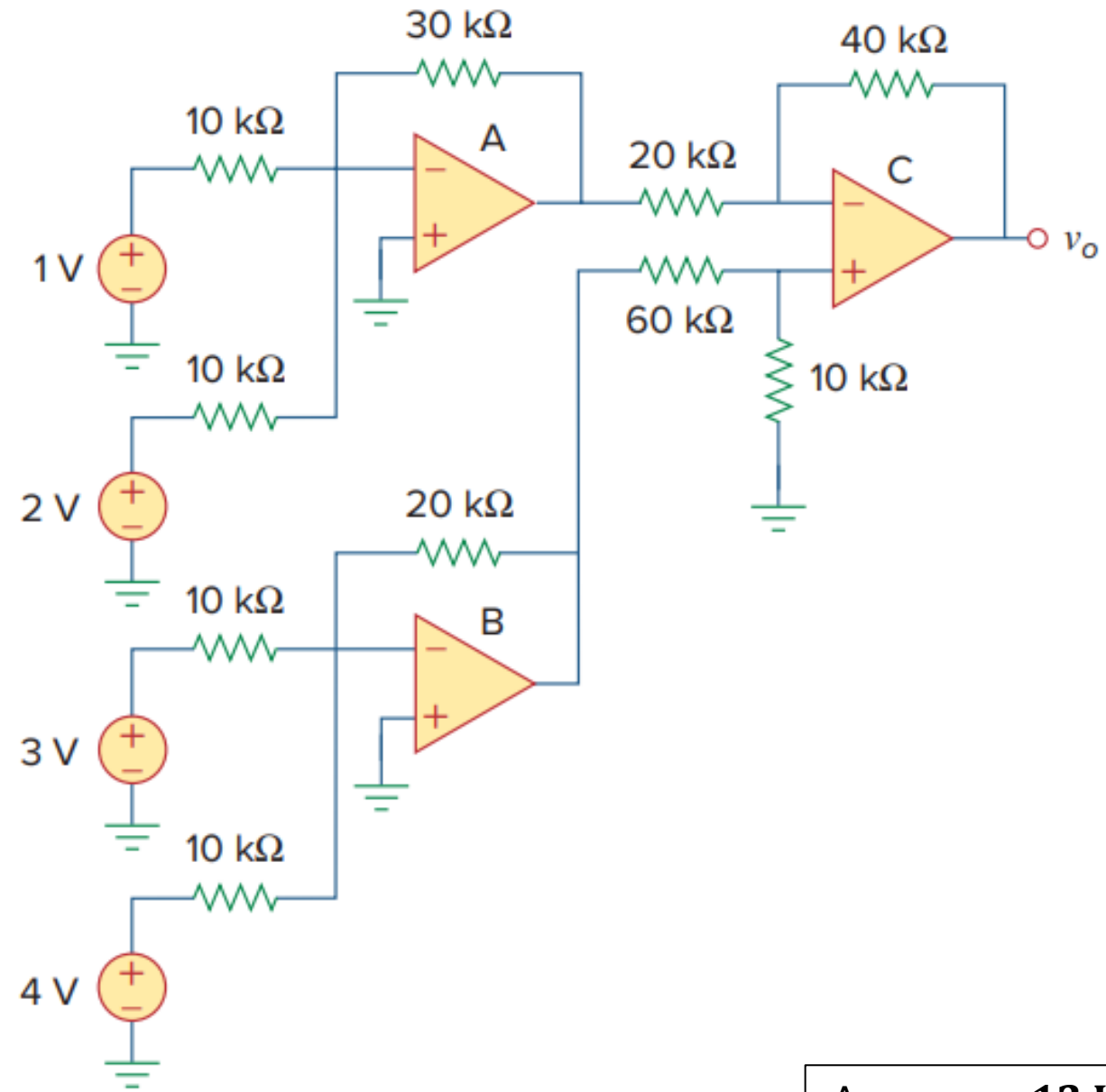
- Determine the value of v_2 in order to make $v_o = -16.5\text{ V}$.



Ans: $v_2 = 10.6\text{ V}$

Problem 7

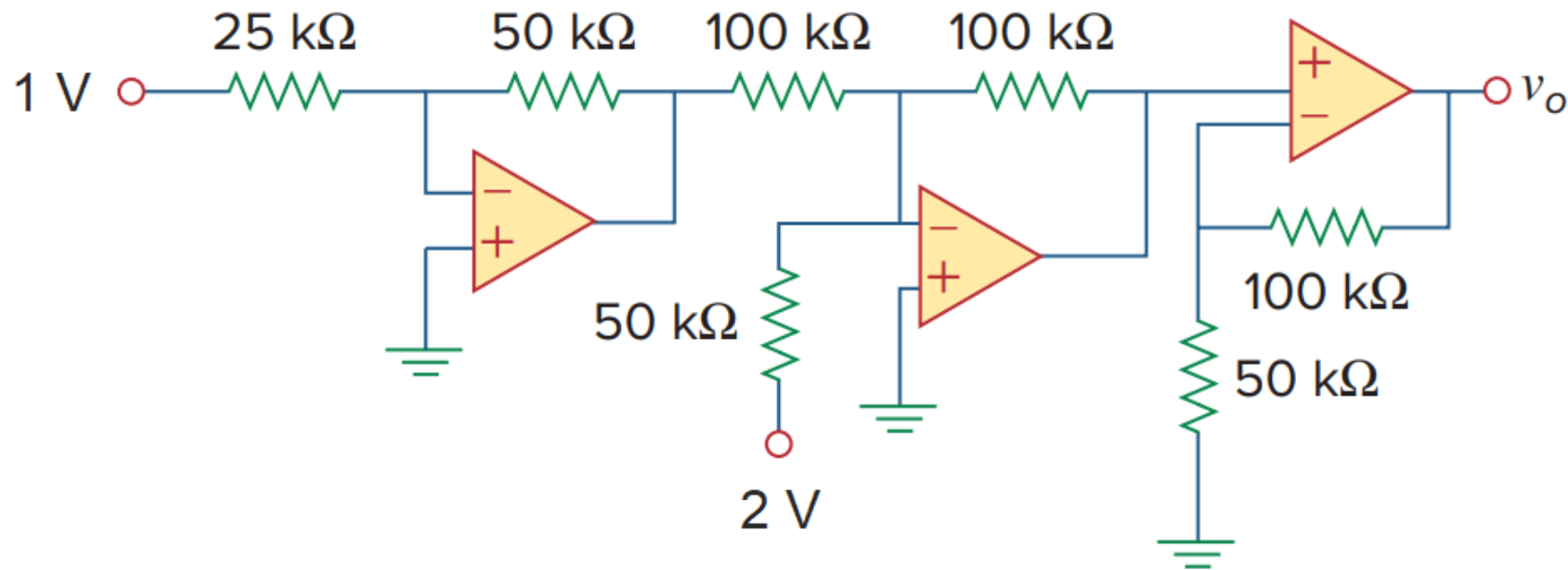
- Determine v_o from the circuit shown.



Ans: $v_o = 12 V$

Problem 8

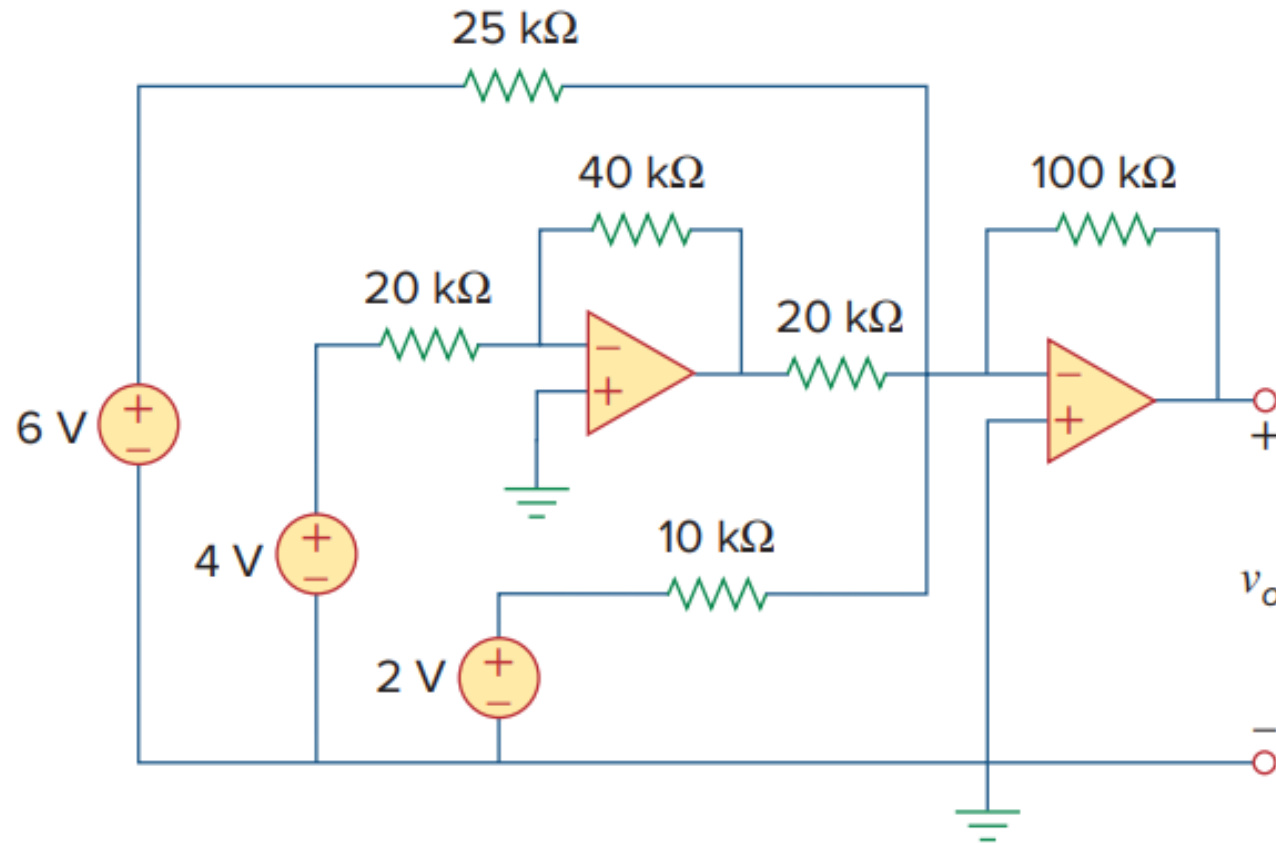
- Determine v_o from the circuit shown below.



Ans: $v_o = -6 \text{ V}$

Problem 9

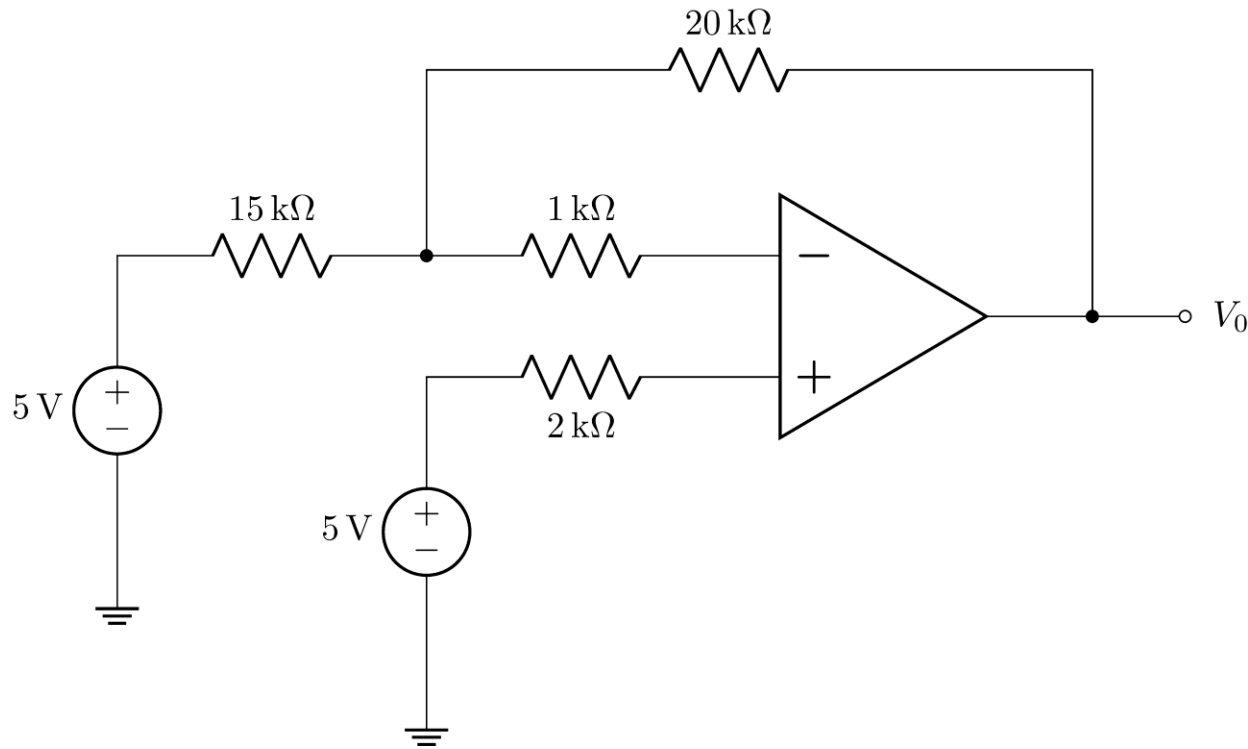
- Determine v_o from the circuit shown below.



Ans: $v_o = -4\text{ V}$

Problem 10

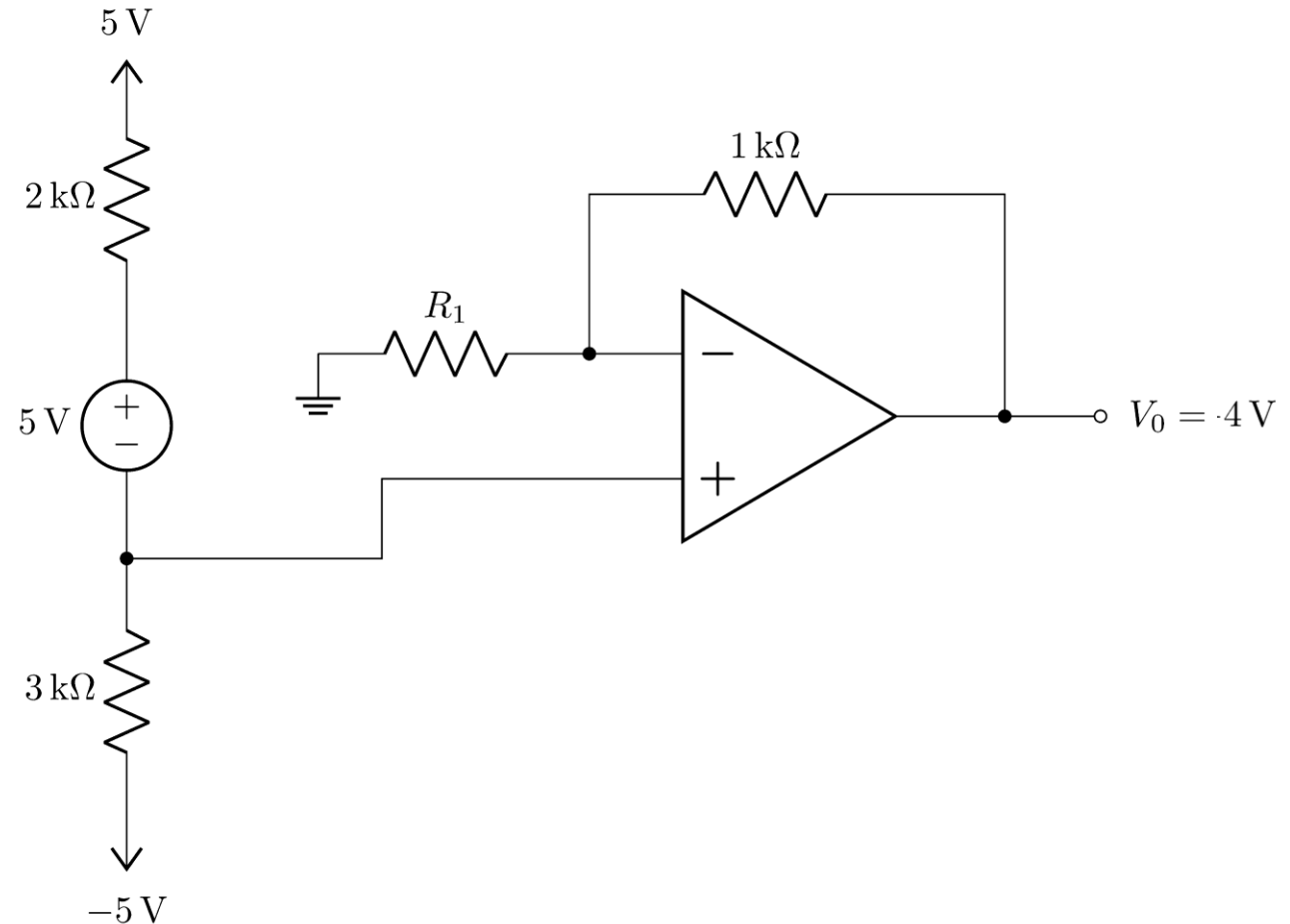
- Determine V_o for the circuit shown below.



Ans: $V_o = 5V$

Problem 11

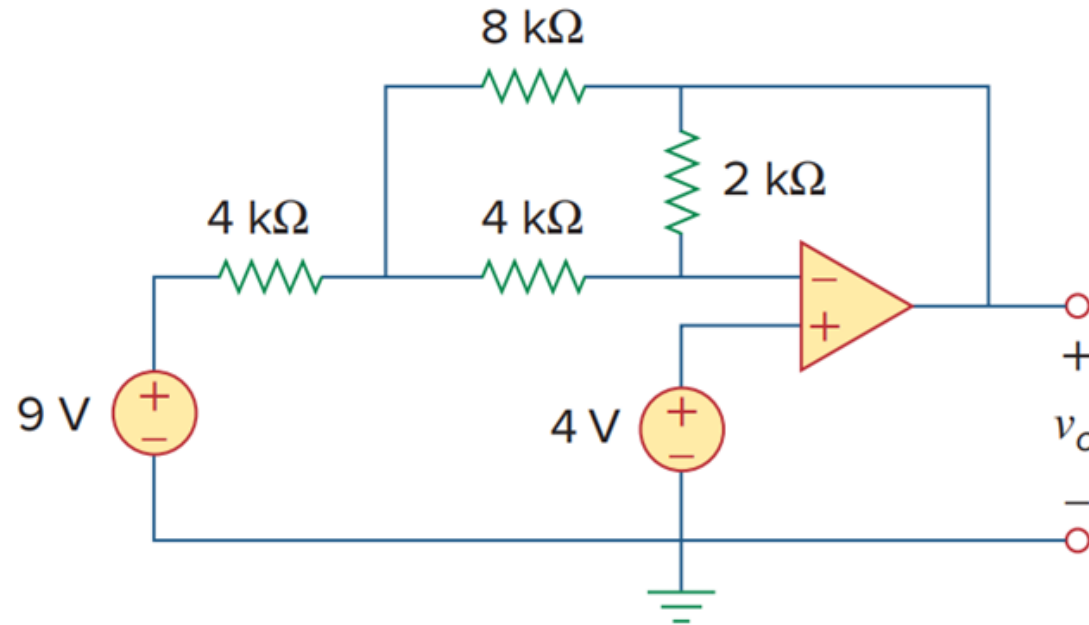
- Determine R_1 for the circuit shown below.



Ans: $R_1 = 1\text{ k}\Omega$

Problem 12**

- Determine v_o for the circuits shown below. [Hint: avoid KCL at V_o as op-amp's output current is not known]

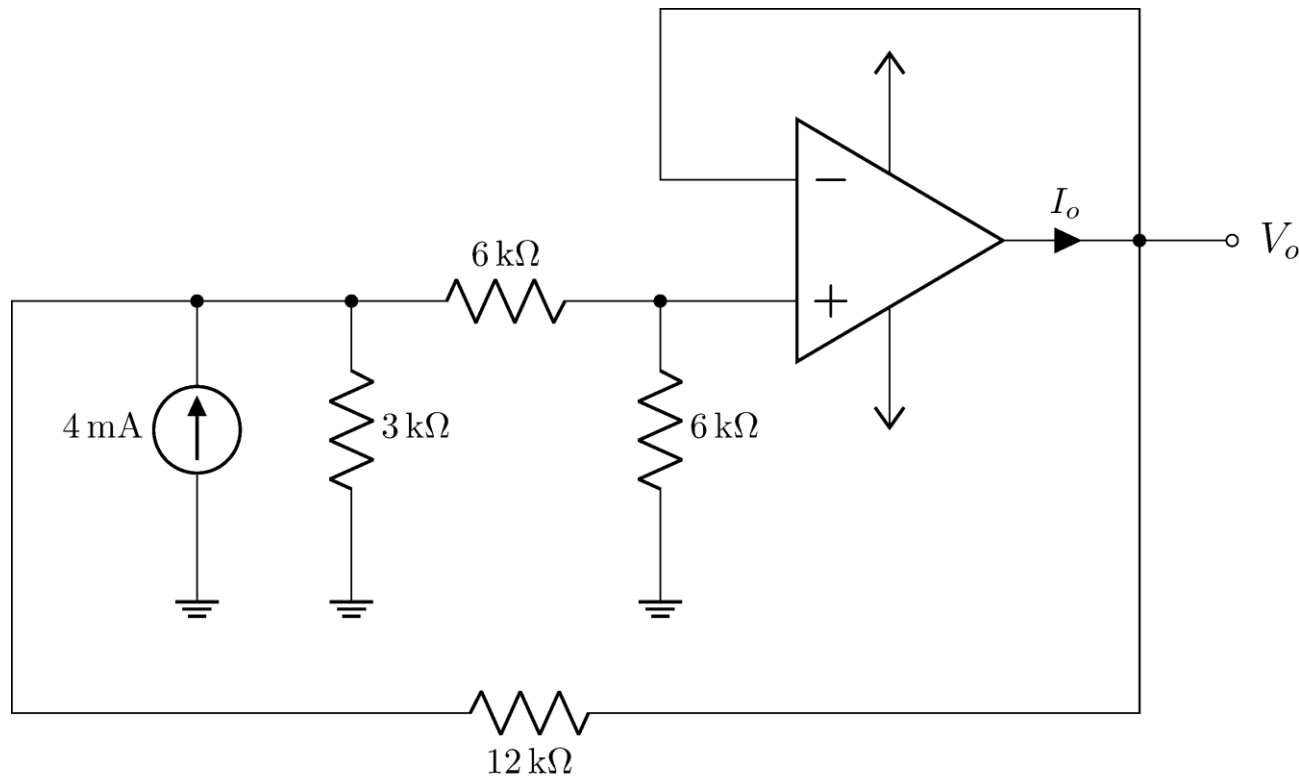


Ans: (a) $v_o = 3.09 V$

Problem 13

- Determine V_o and I_o for the circuit shown below.

[Hint: avoid KCL at V_o as the op-amp's output current is not known]

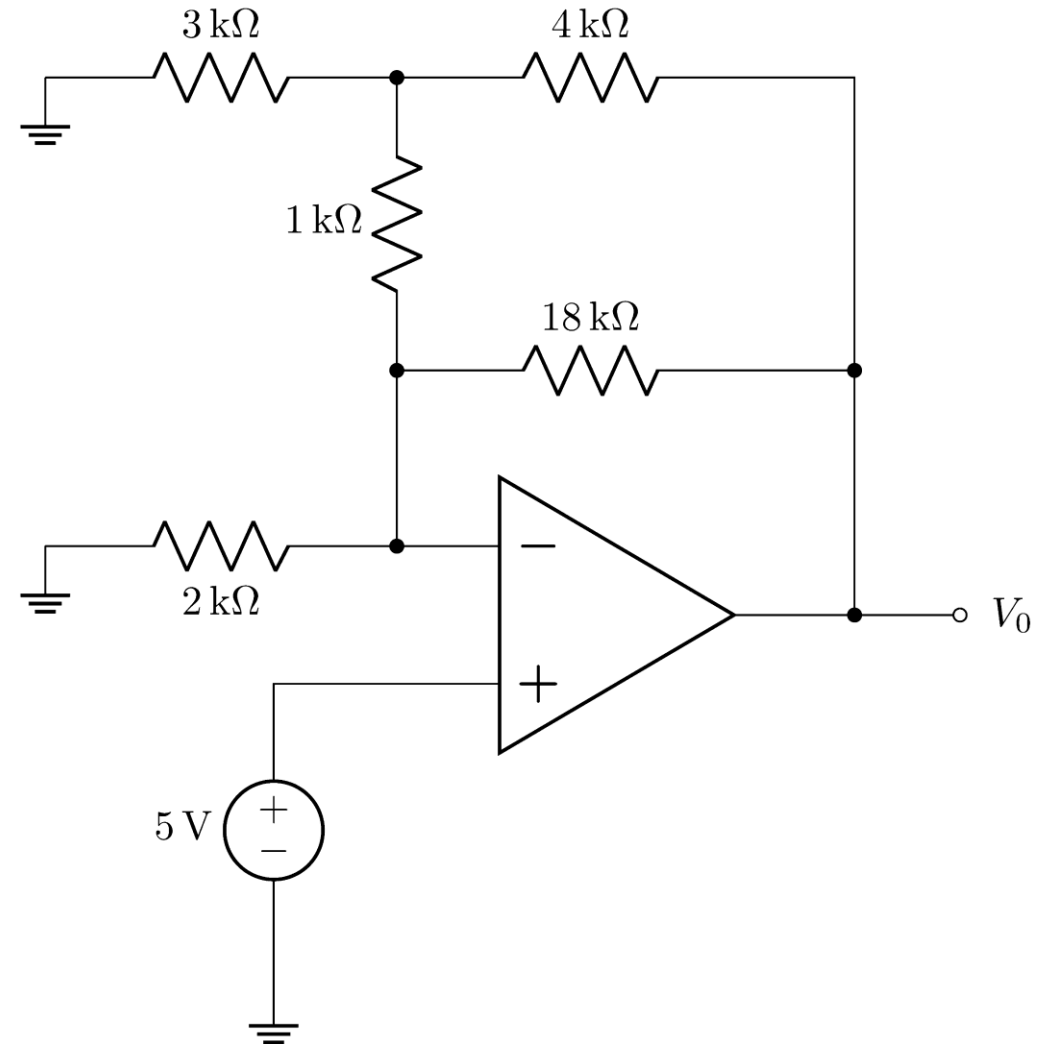


Ans: $V_o = 4.36 \text{ V}$

Problem 14**

- Determine V_o and I_o for the circuit shown below.

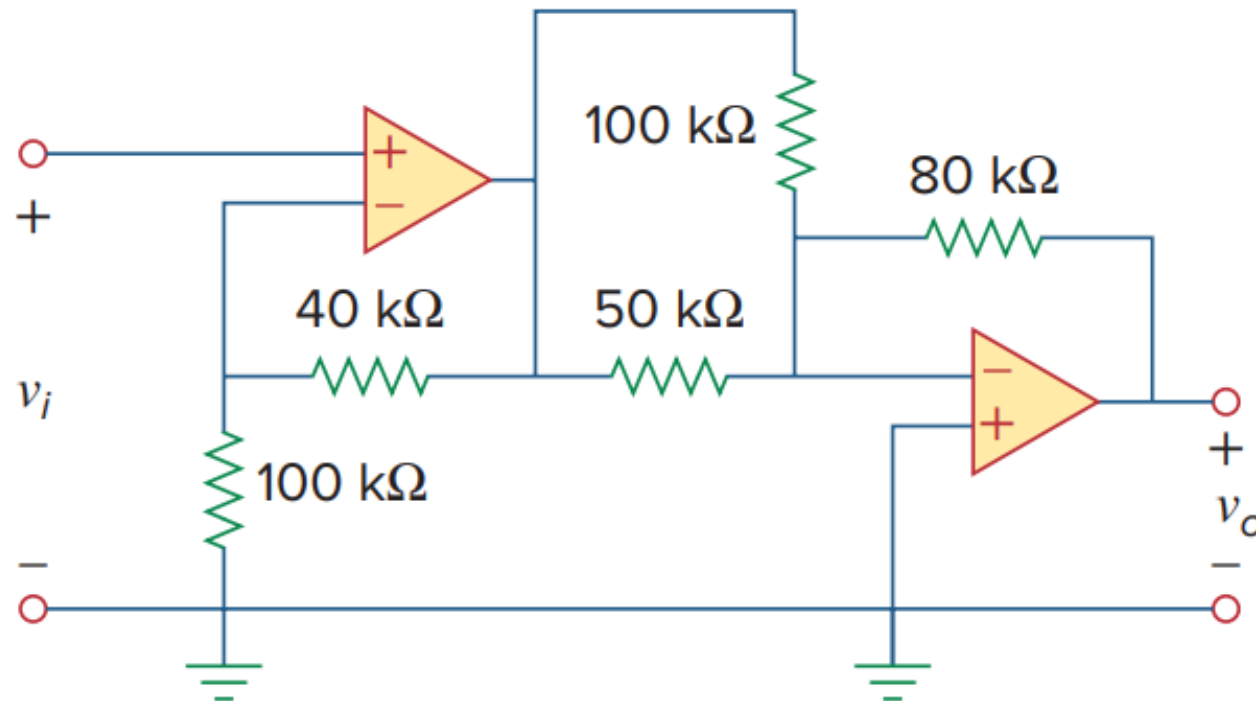
[Hint: avoid KCL at V_o as the op-amp's output current is not known]



Ans: $V_o = 10.4 \text{ V}$

Problem 15**

- Determine the gain (v_o/v_i) from the circuit shown below.

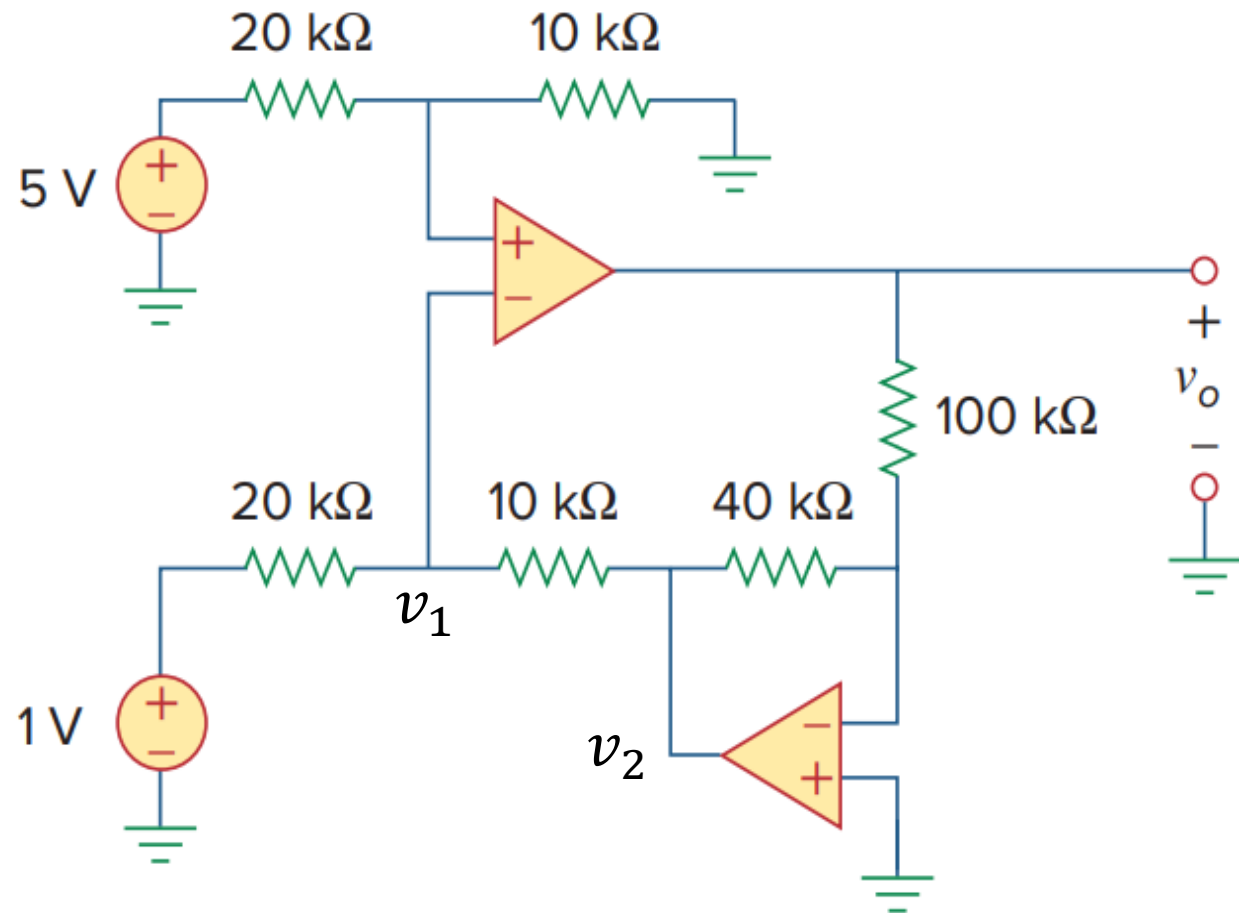


Ans: $v_o/v_i = 3.36$

Problem 16**

- I. Determine v_1 .
- II. Determine v_2 and v_o .

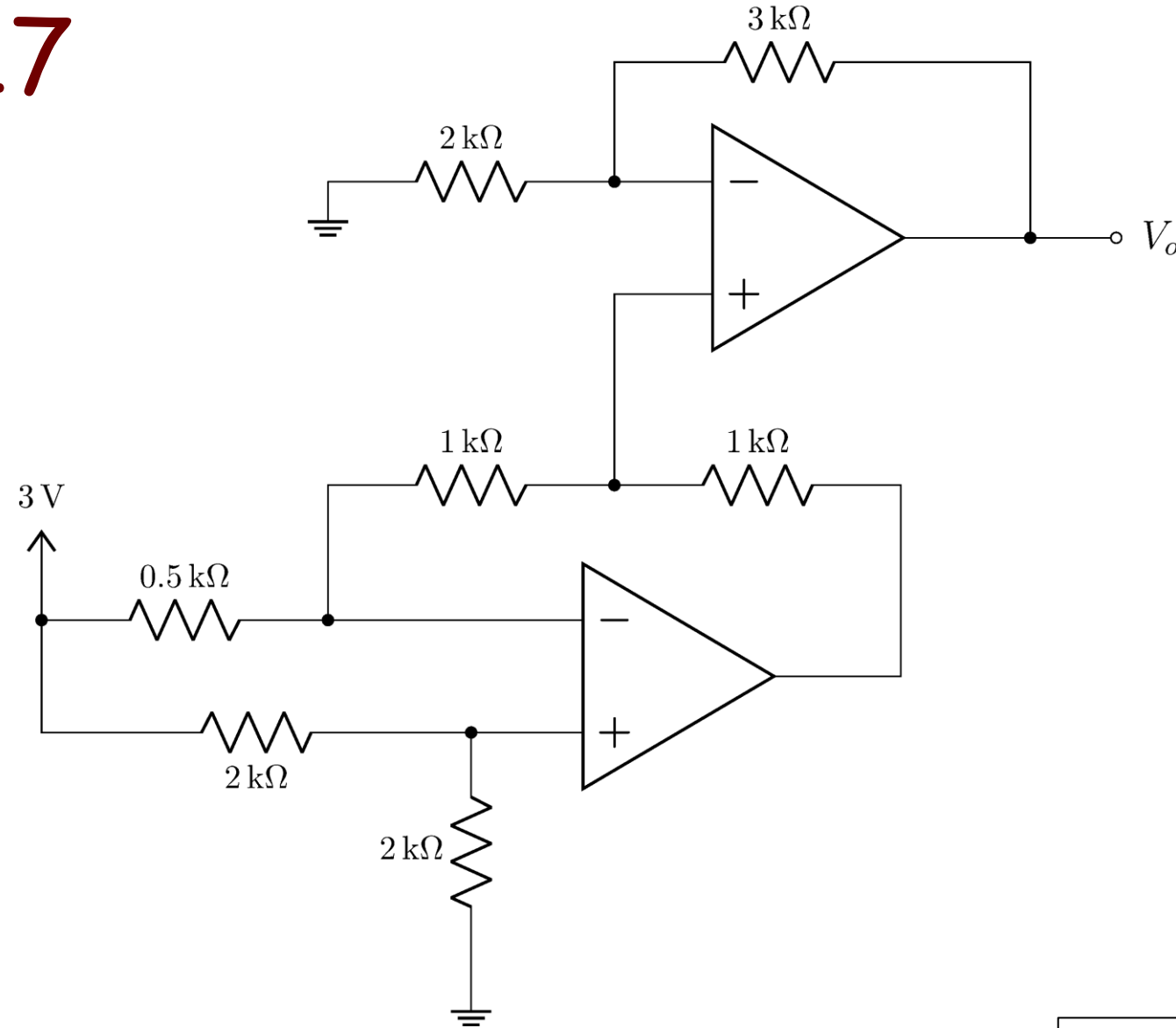
[Hint: avoid KCL at v_o and v_2 as the op-amps' output currents are not known]



Ans: $v_1 = \frac{5}{3} \text{ V}$, $v_2 = 2 \text{ V}$, $v_o = -5 \text{ V}$

Problem 17

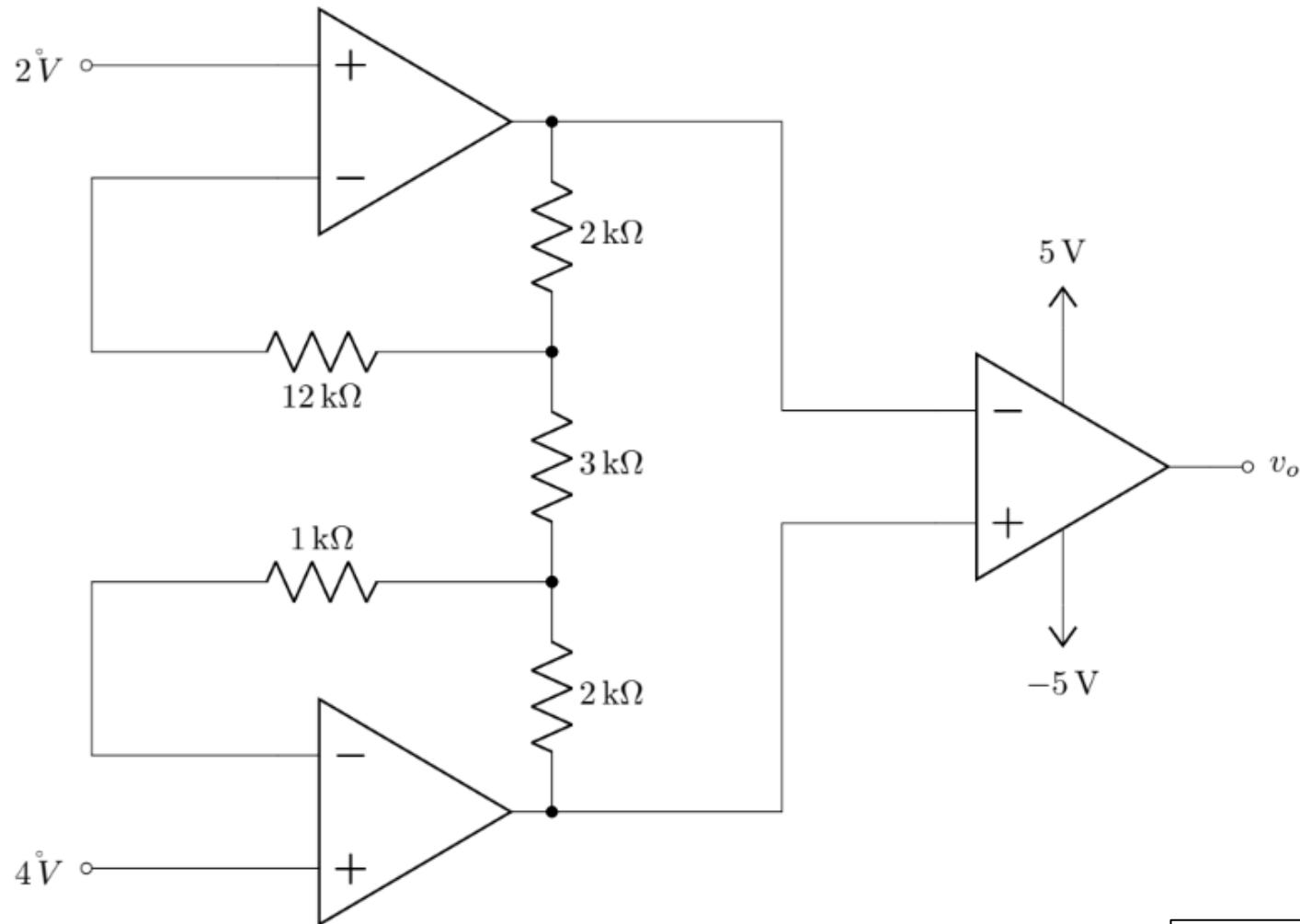
- Determine V_o .



Ans: $V_o = -3.75 \text{ V}$

Problem 18**

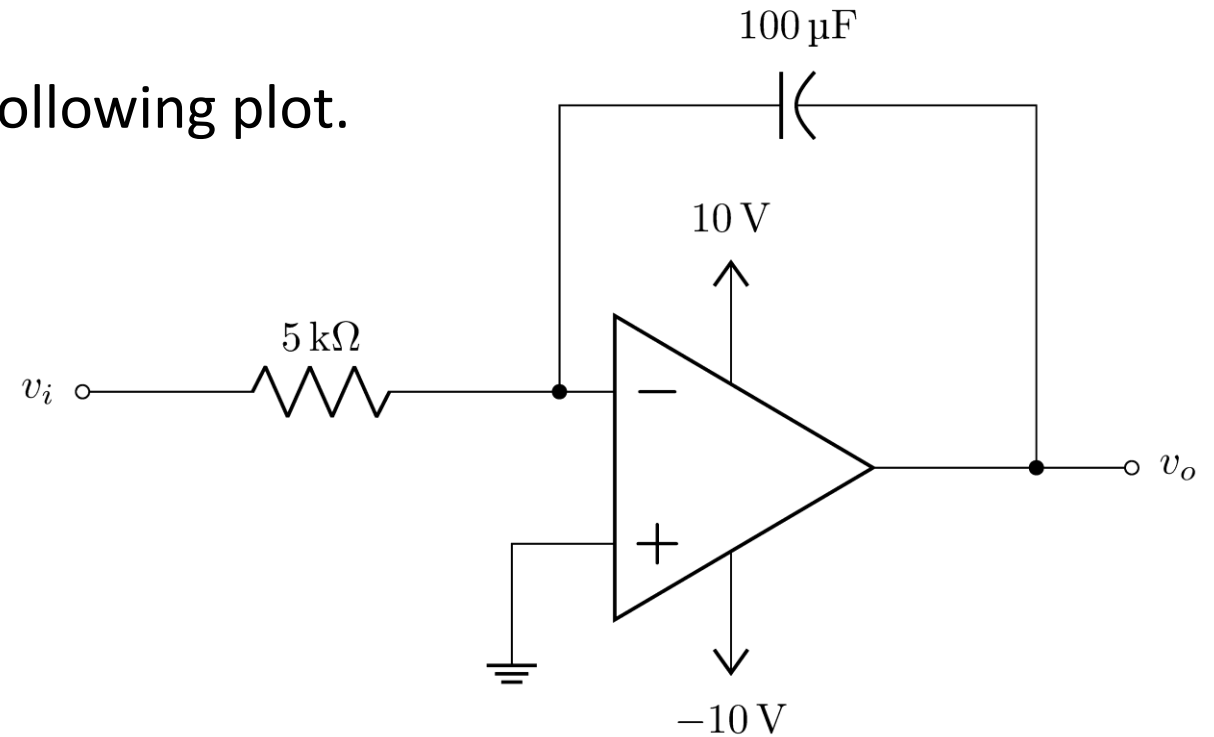
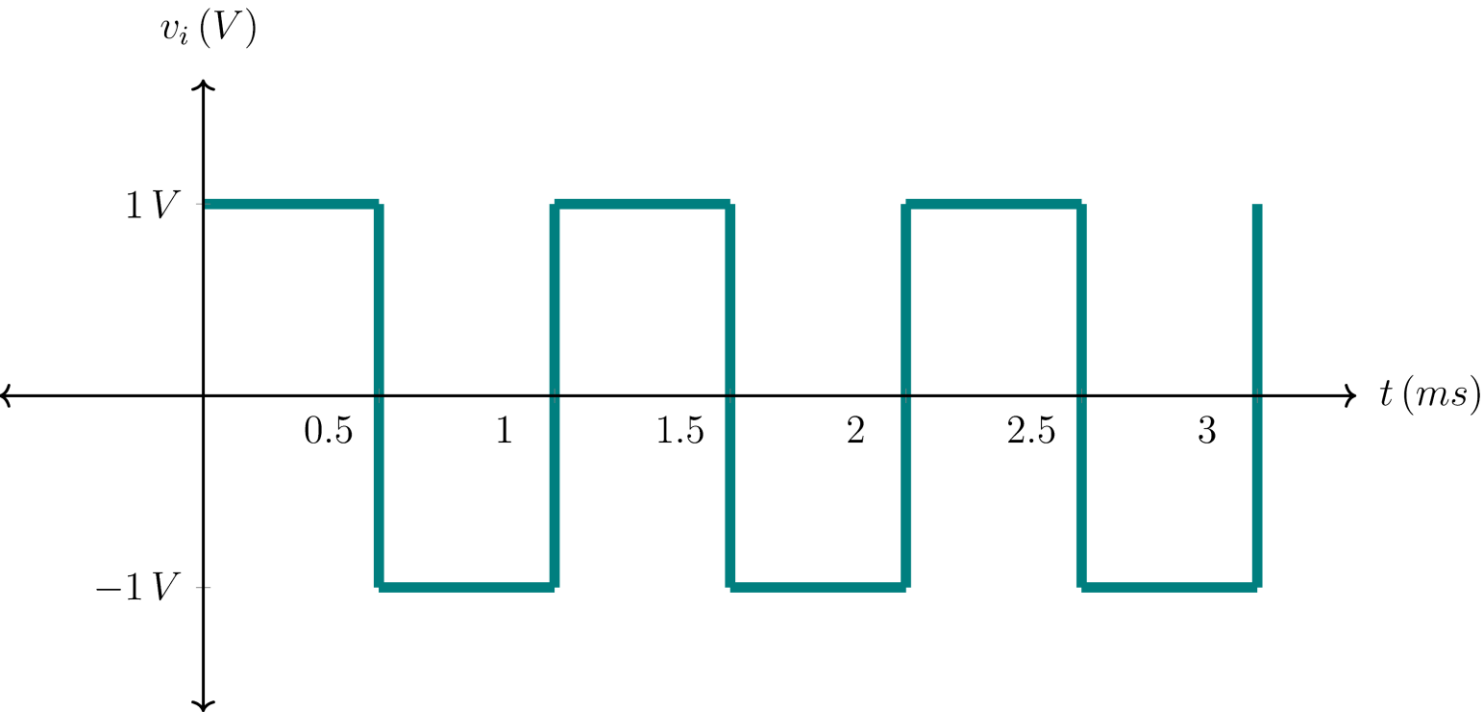
- Determine v_o .



Ans: $v_o = 5V$

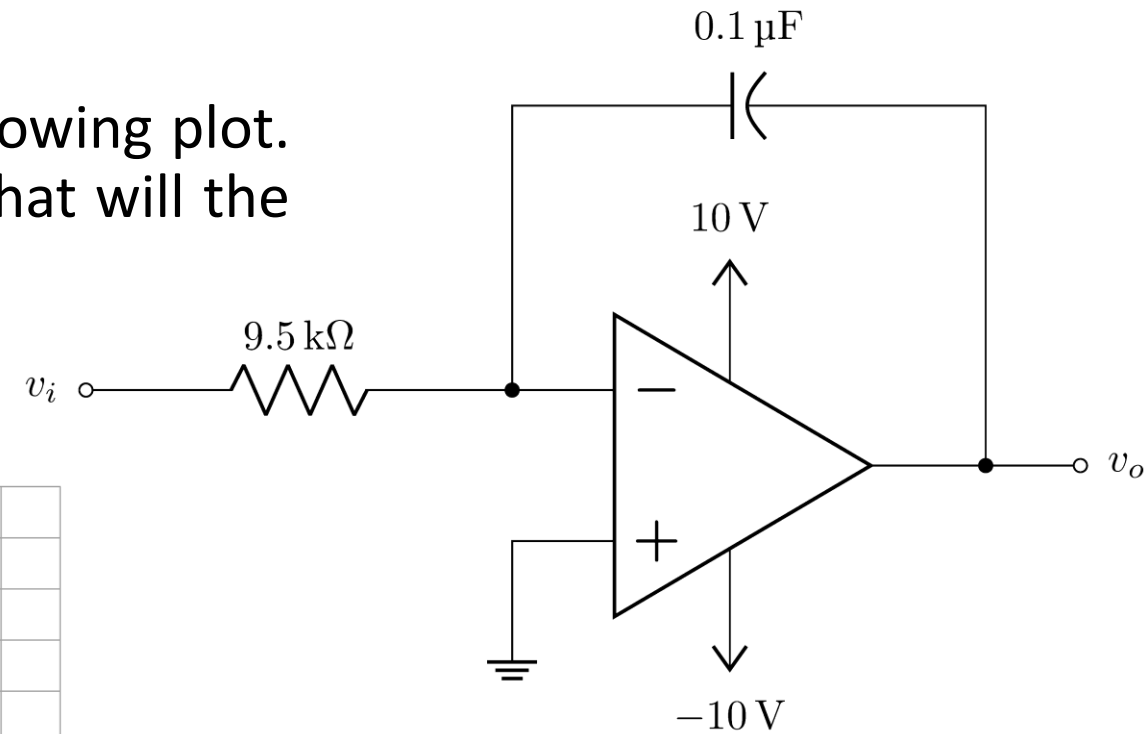
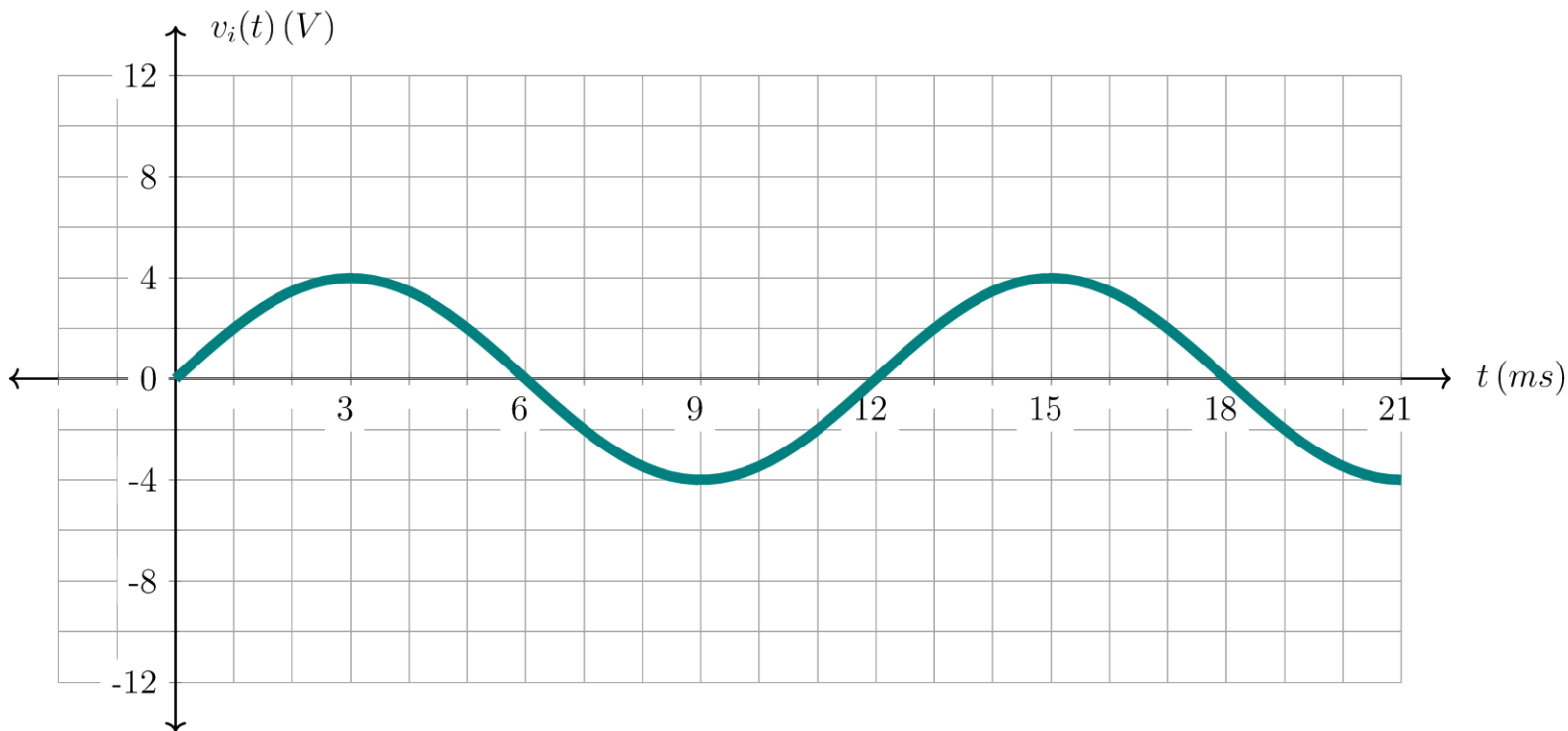
Problem 19

- Sketch v_o vs. t , if v_i is as shown in the following plot.



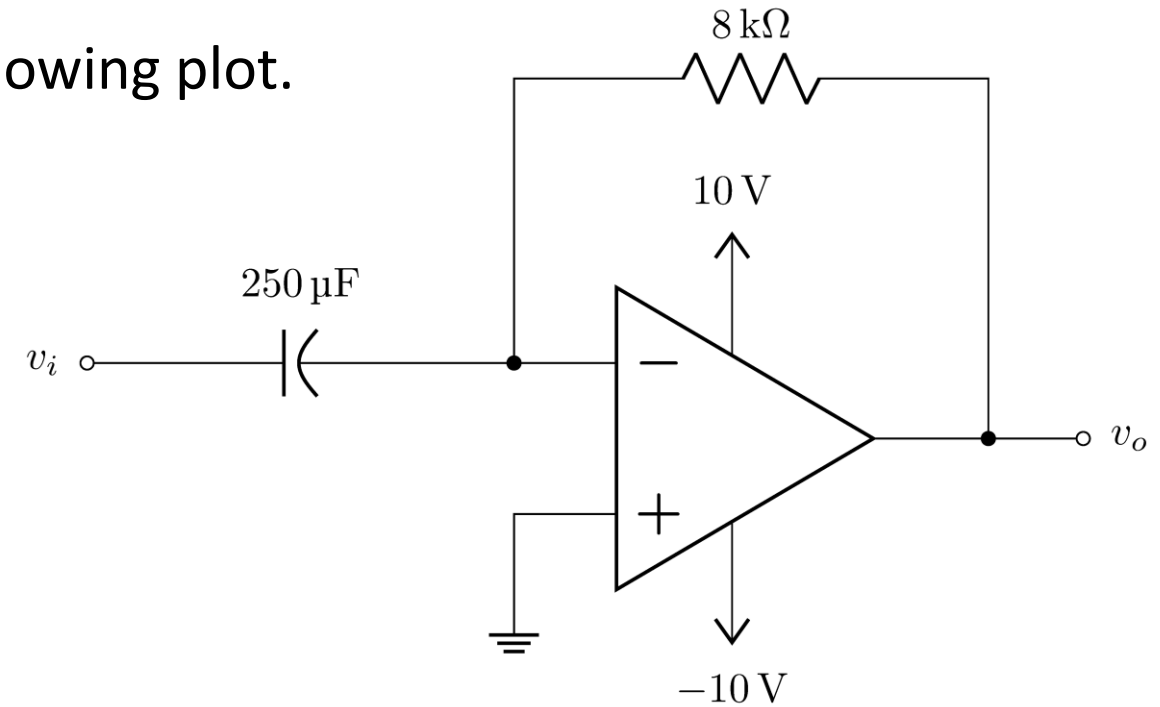
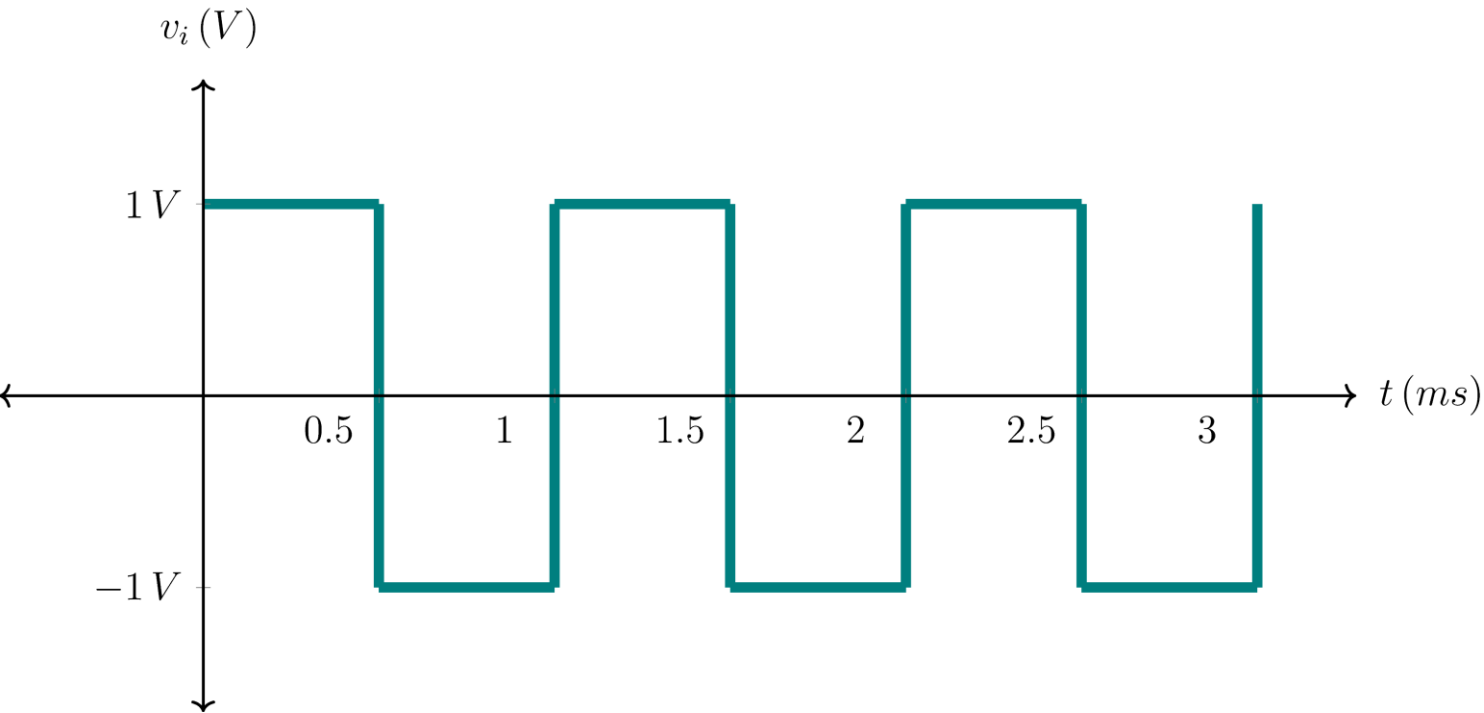
Problem 20

- Sketch v_o vs. t , if v_i is as shown in the following plot. If the bias voltages are $+6\text{ V}$ and -6 V , what will the graph look like.



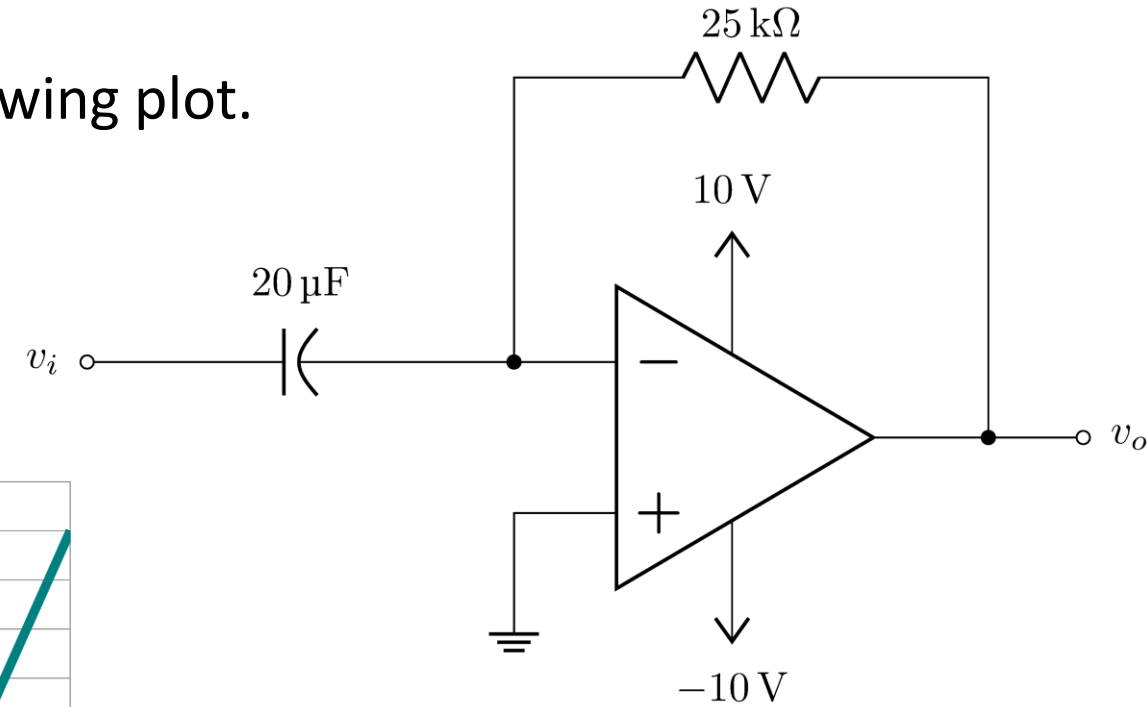
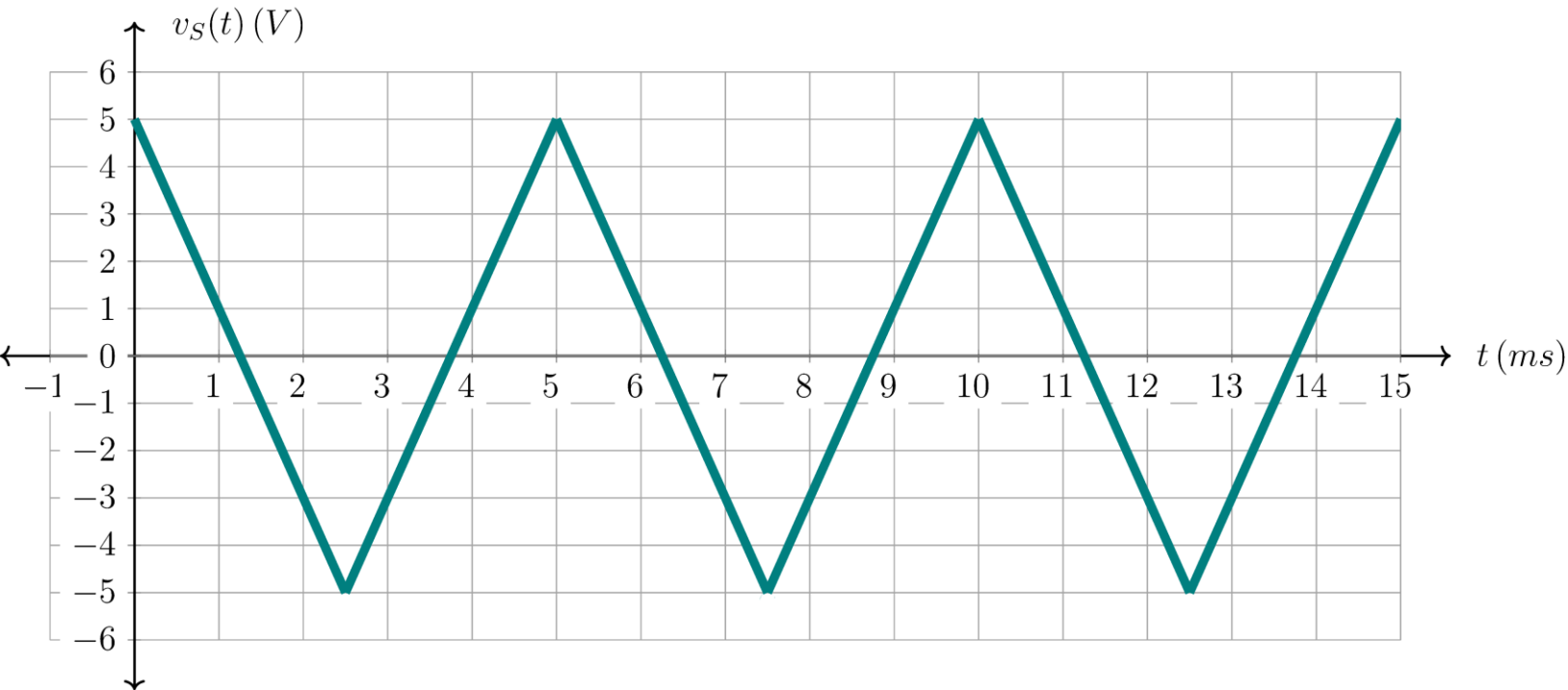
Problem 21

- Sketch v_o vs. t , if v_i is as shown in the following plot.



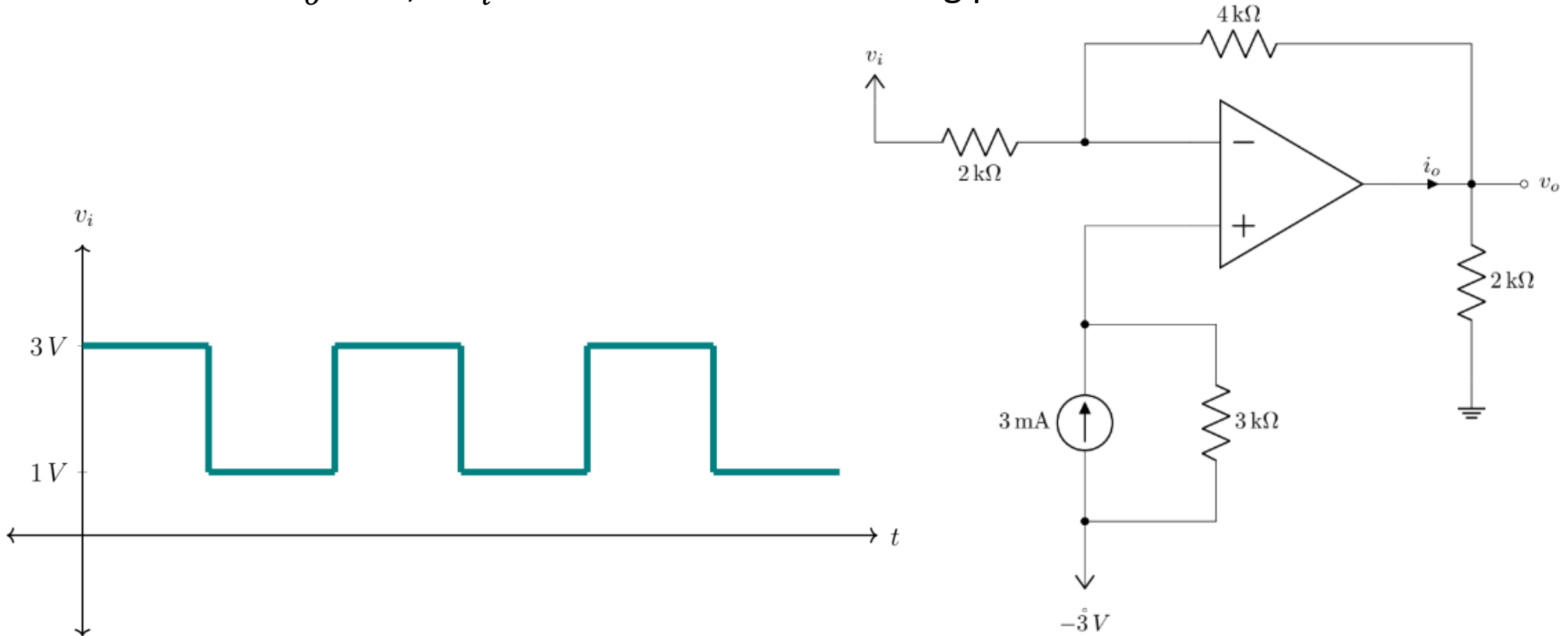
Problem 22

- Sketch v_o vs. t , if v_i is as shown in the following plot.



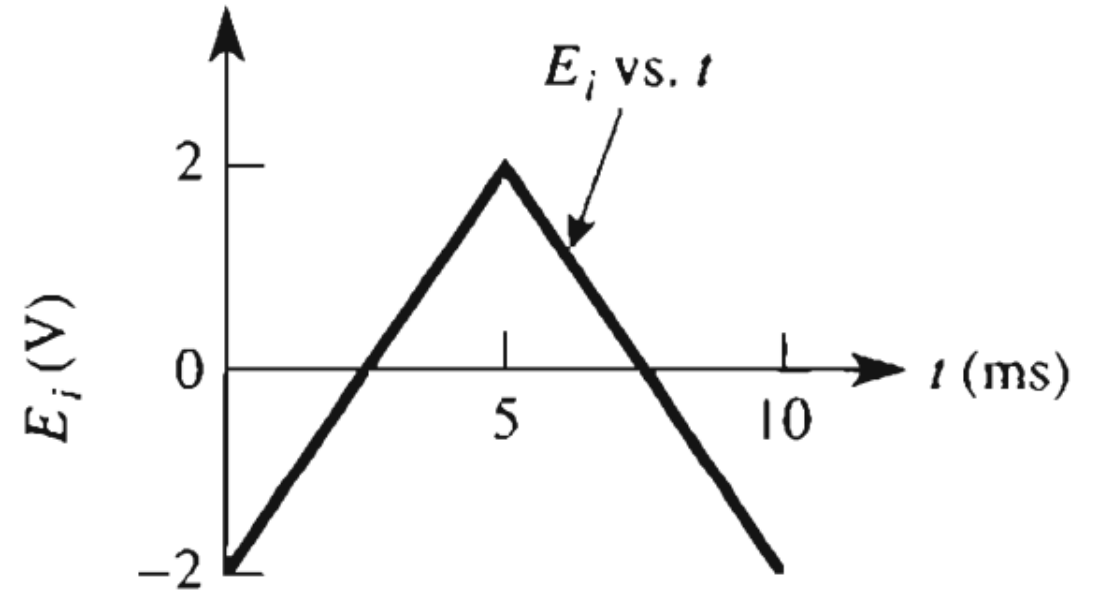
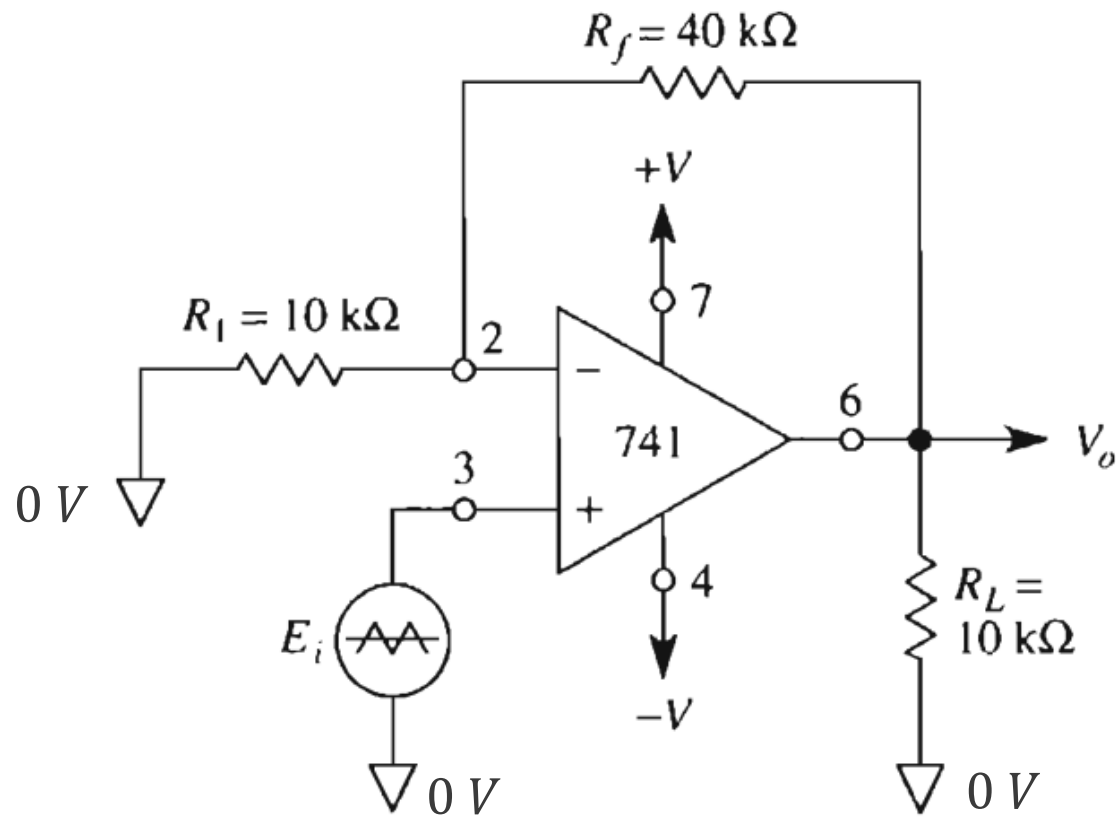
Problem 23

- Sketch i_o vs. t , if v_i is as shown in the following plot.



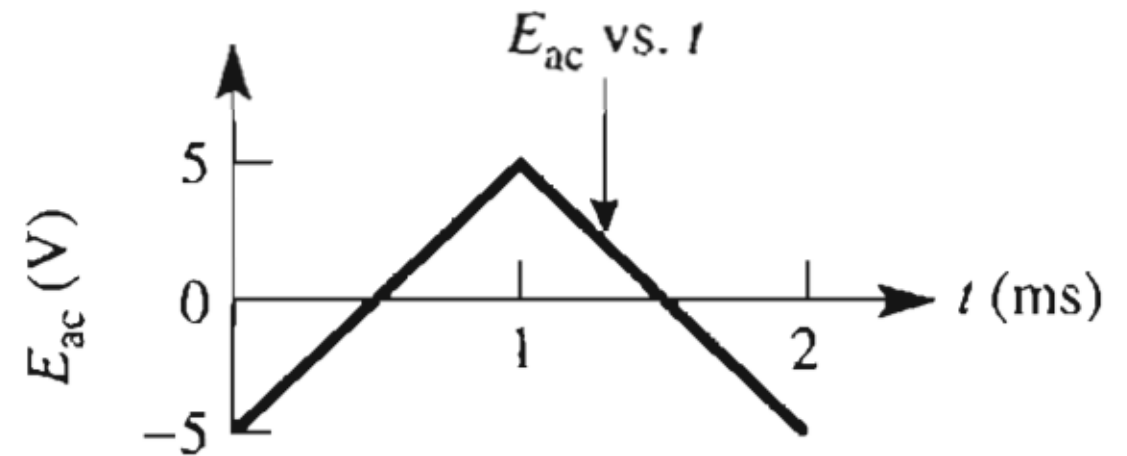
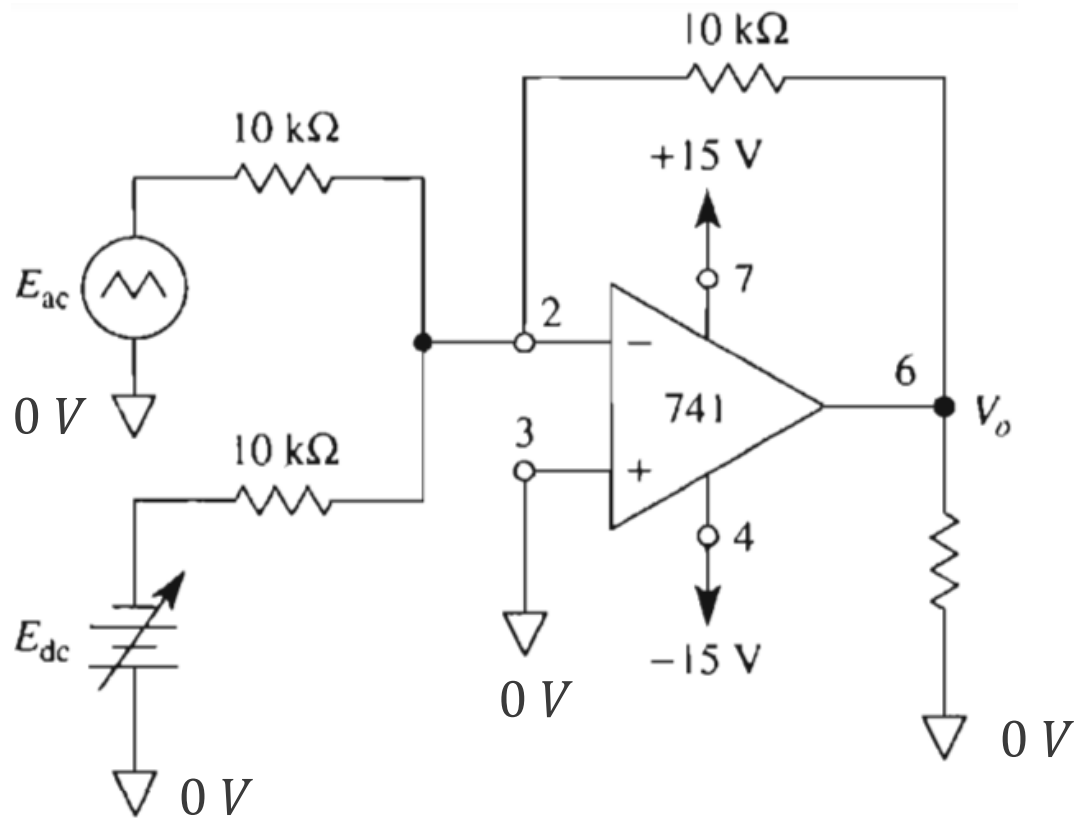
Problem 24

- Sketch V_o vs. t and V_o vs. E_i .



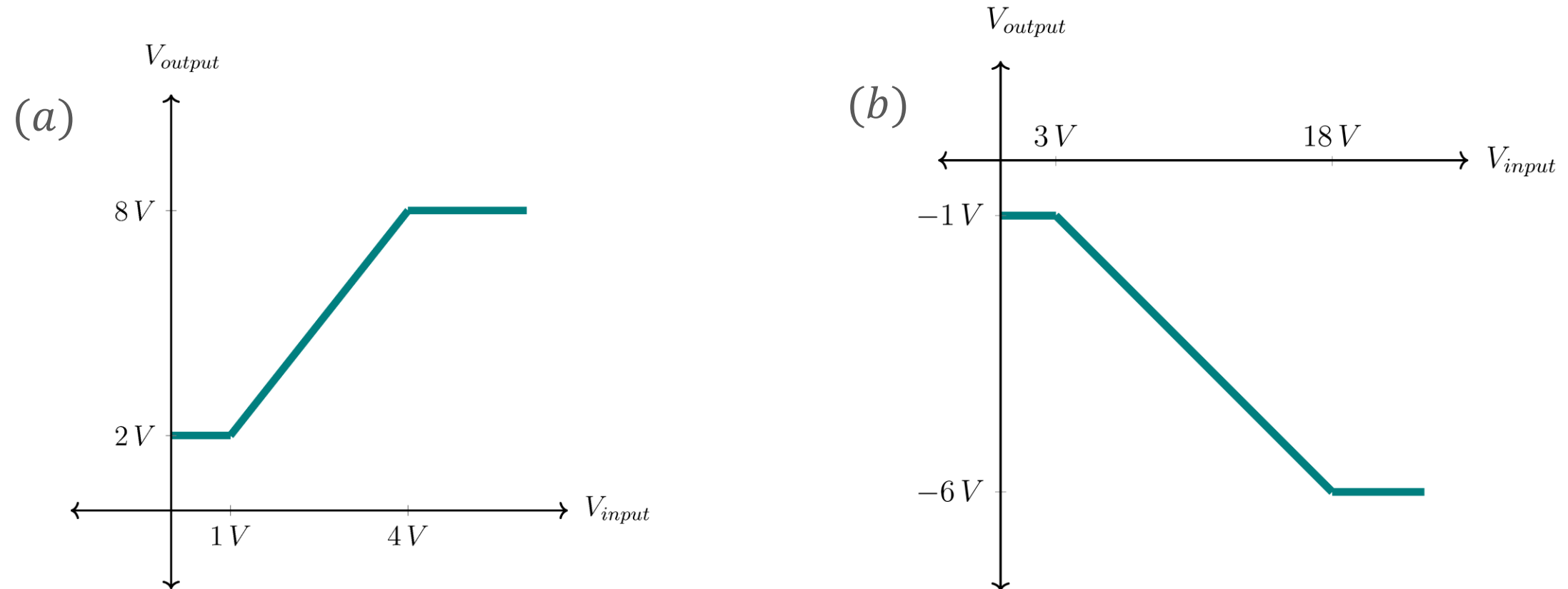
Problem 25

- Sketch V_o vs. t and V_o vs. E_{ac} , if $E_{dc} = -5$ V.



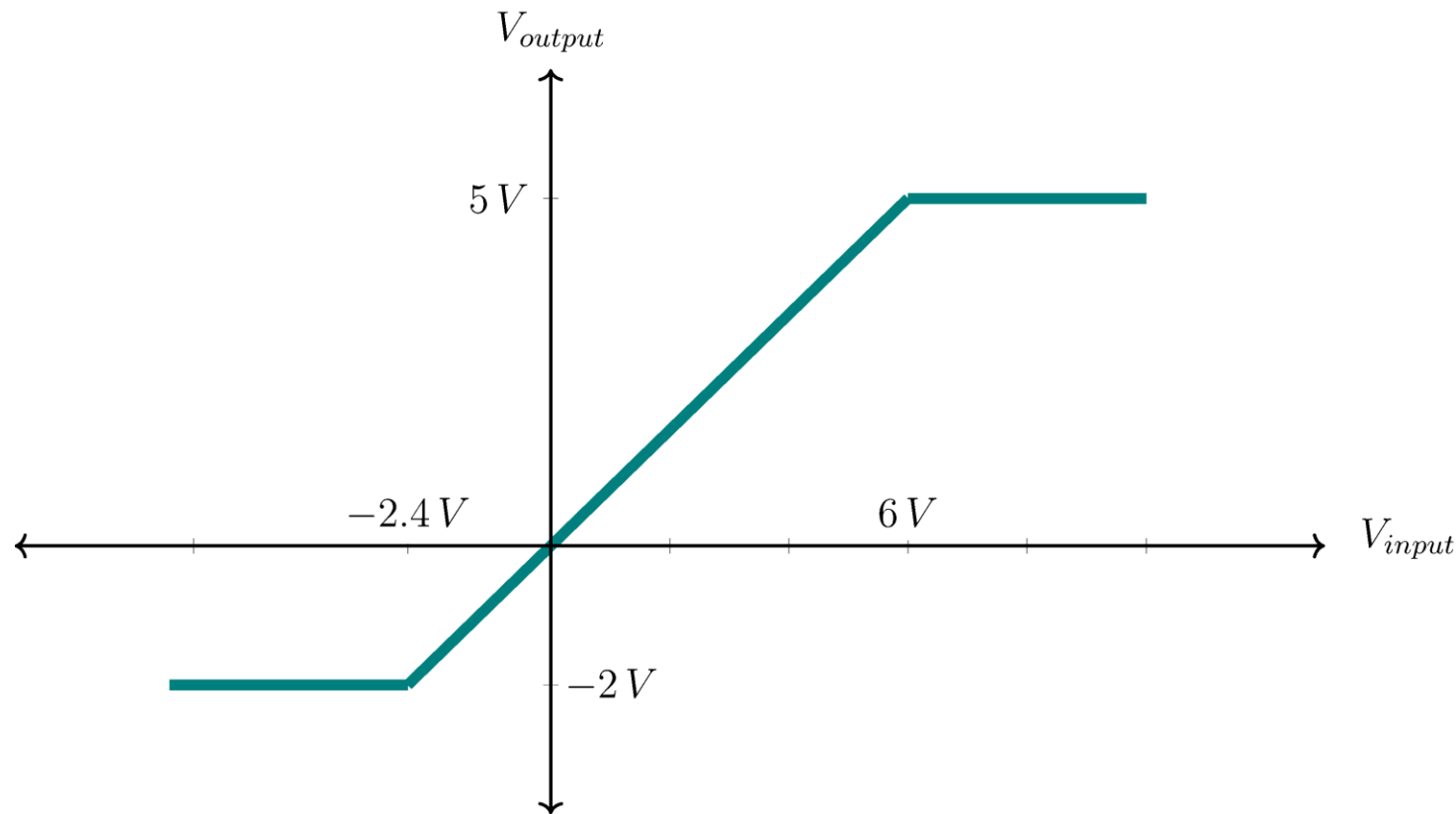
Problem 26

- Design a circuit with a single ideal op-amp for each of the VTC plots shown below. V_{output} and V_{input} are the output voltage and the input voltage respectively.



Problem 27

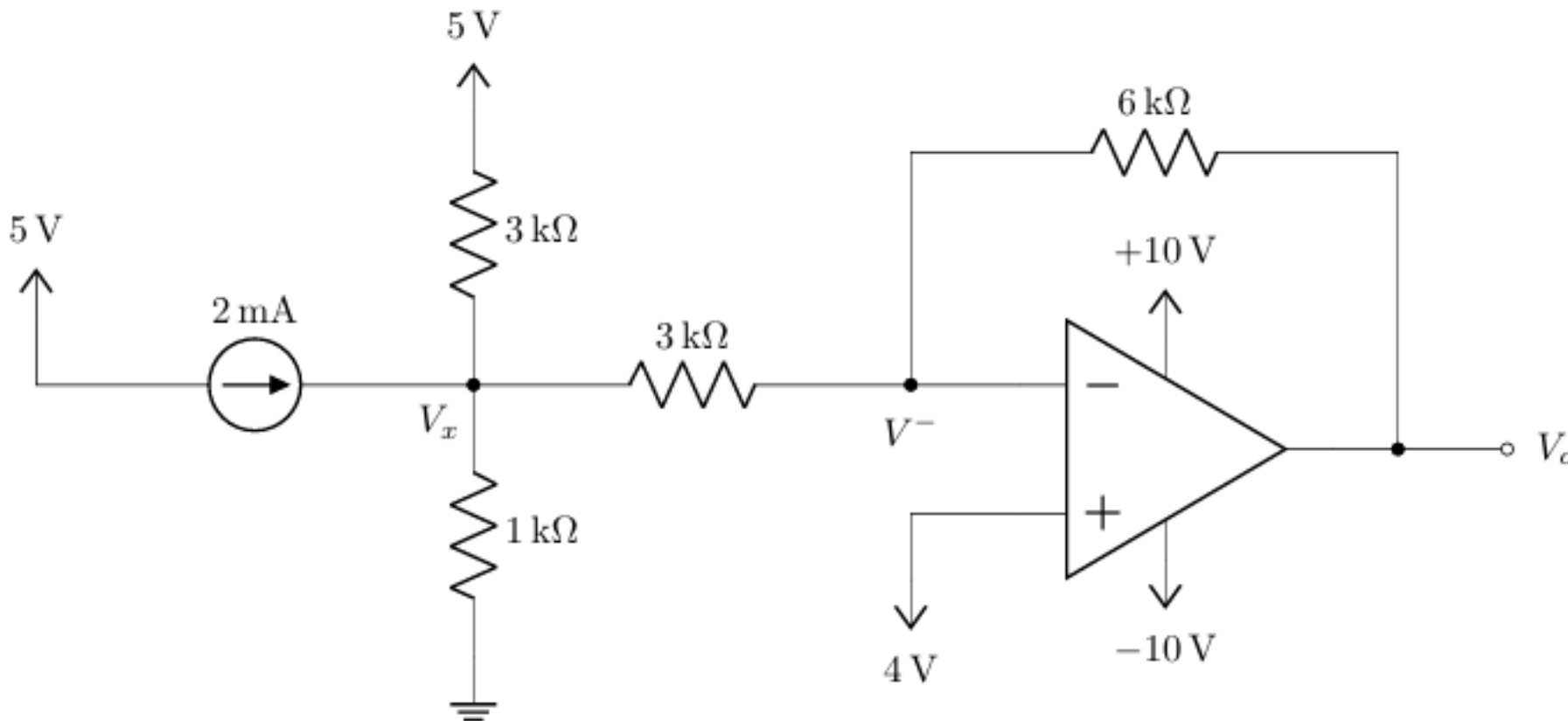
- Design a circuit with a single ideal op-amp for the VTC plots shown below. V_{output} and V_{input} are the output voltage and the input voltage respectively.



Up Next Relevant *Questions* from Past Semesters

Mid — Fall 2025 - 1/2

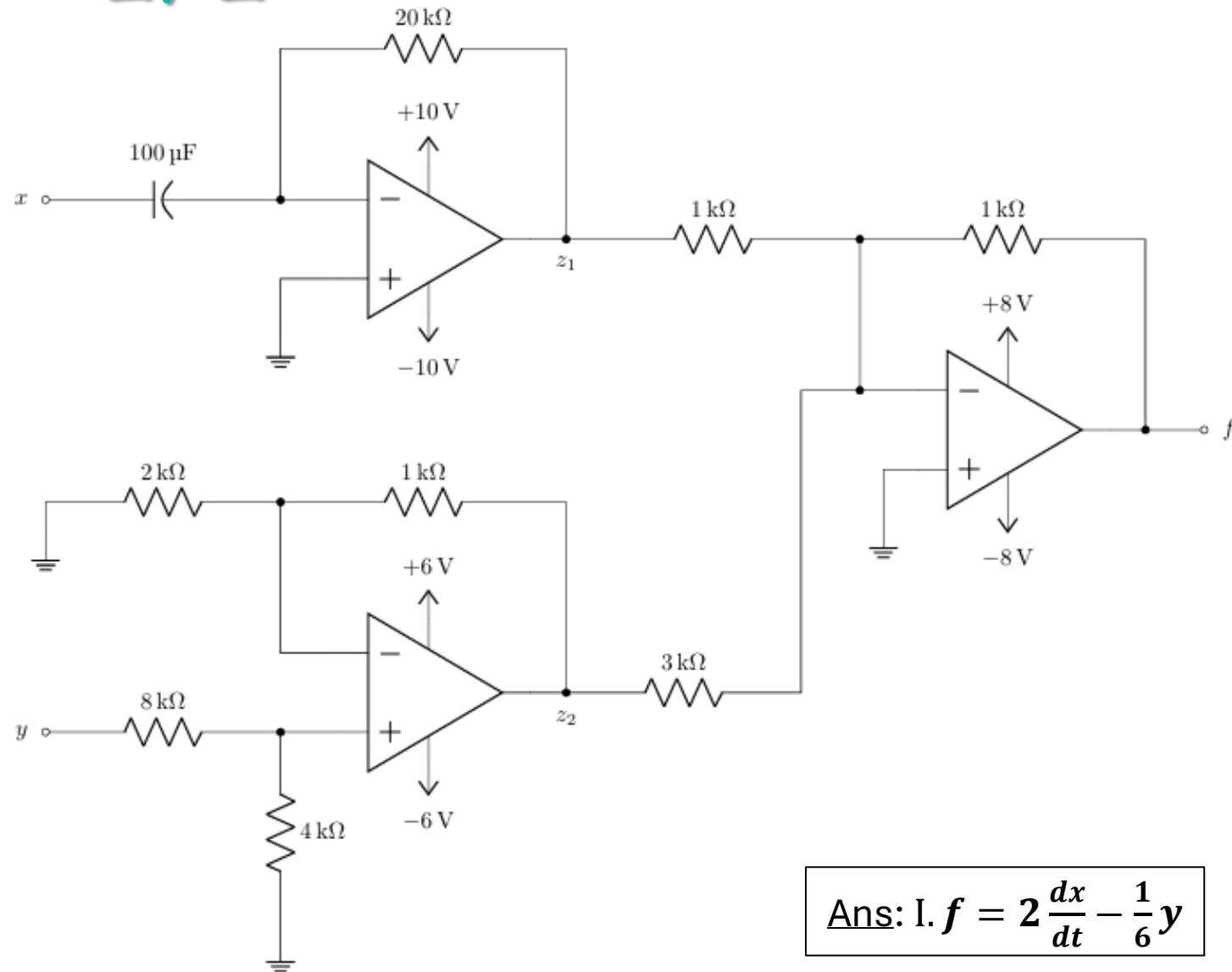
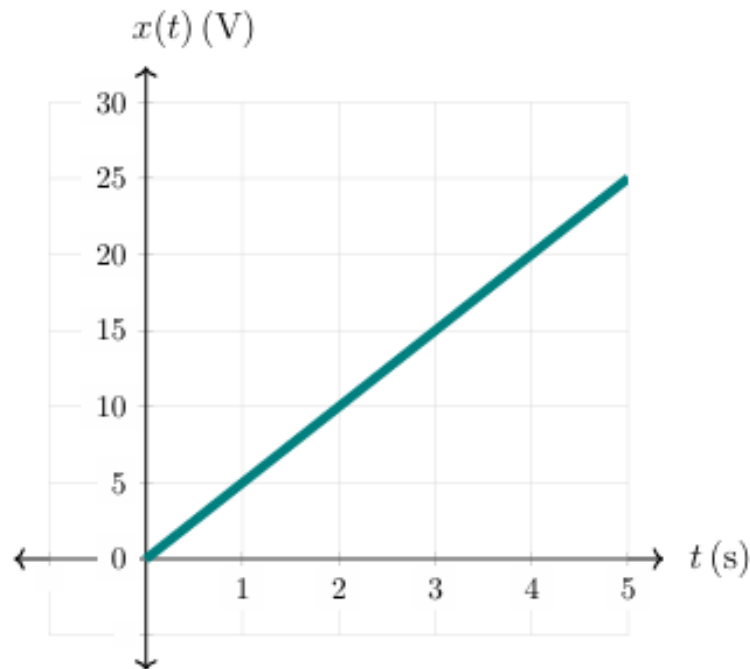
- Determine V_x and V_o in the circuit shown below.



Ans: $V_x = 3\text{ V}, V_o = 6\text{ V}$

Mid — Fall 2025 - 2/2

- I. Express f as a function of variables x and y .
- II. If x is a time-varying voltage as shown in the following plot, sketch the corresponding plot of z_1 .

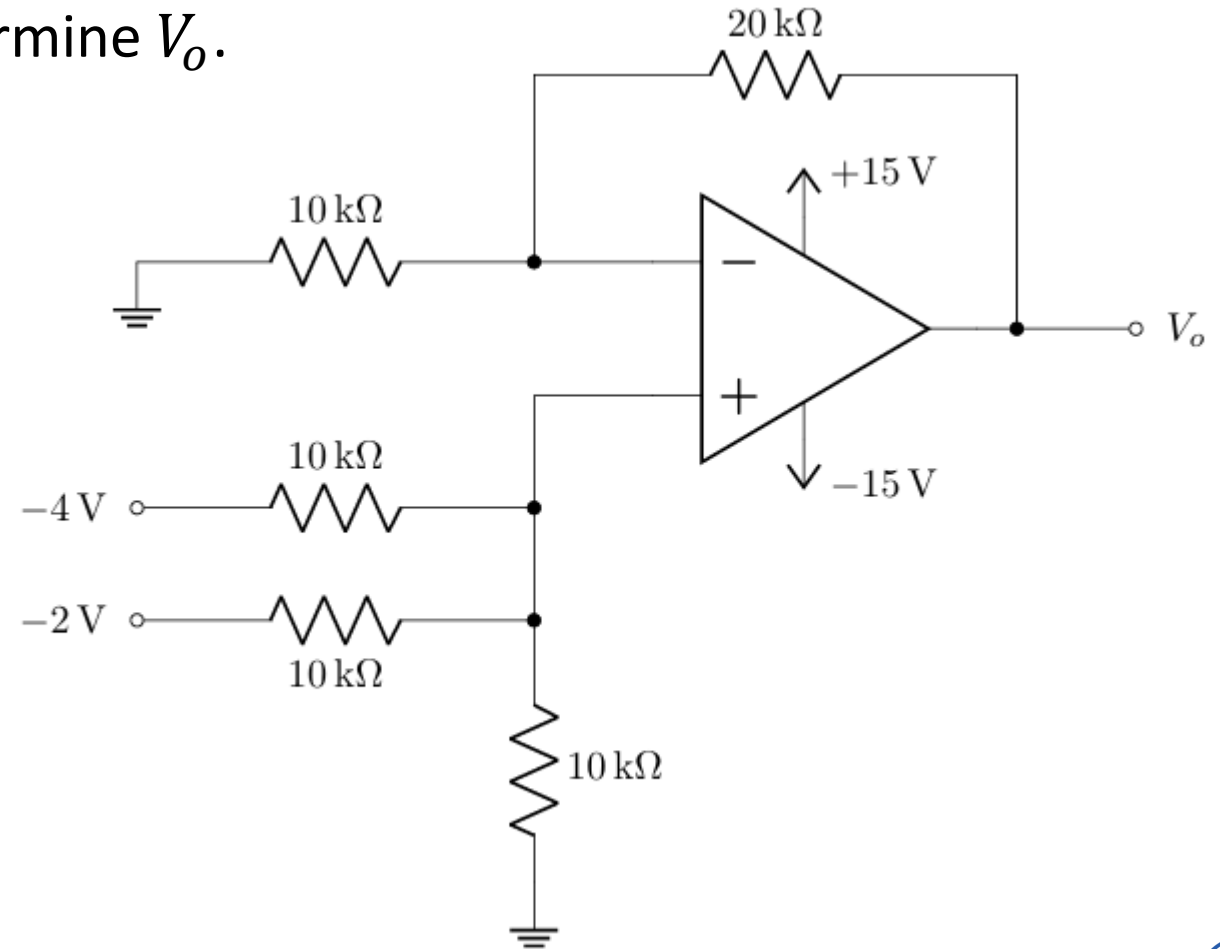


Ans: I. $f = 2 \frac{dx}{dt} - \frac{1}{6} y$



Mid — Summer 2025 - 1/3

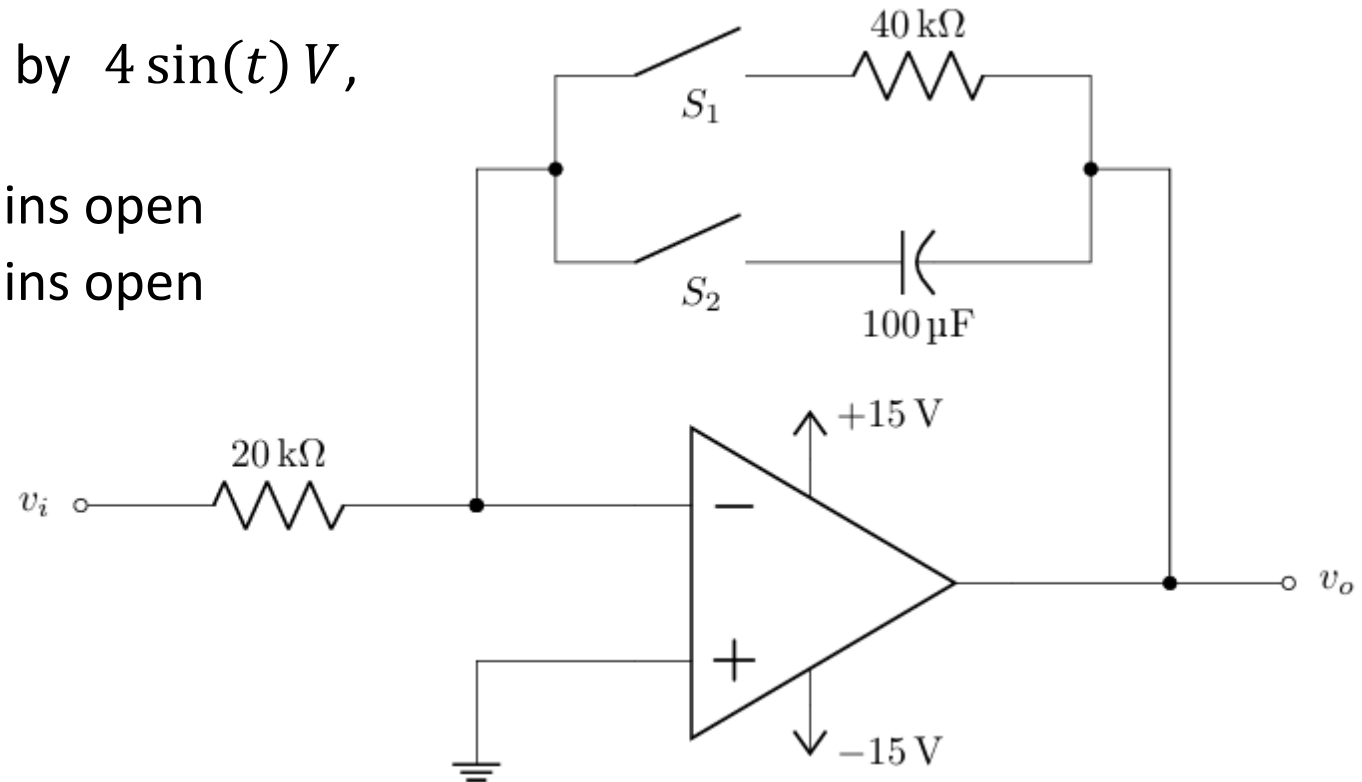
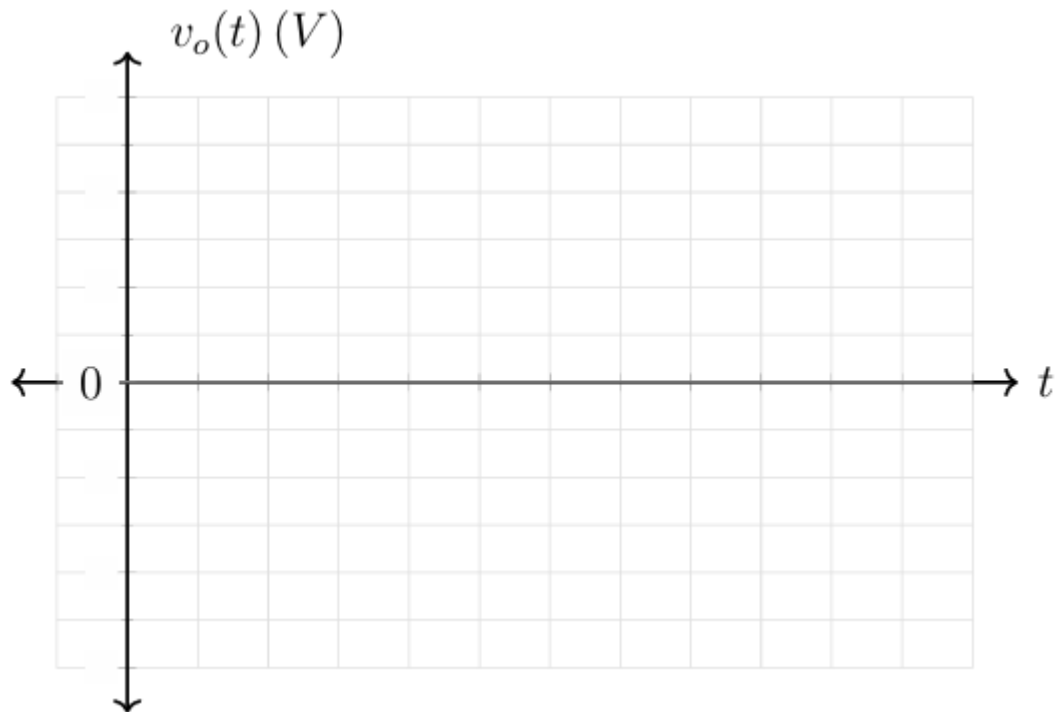
- Analyze the adjacent circuit and determine V_o .



Ans: $V_o = -6\text{ V}$,

Mid — Summer 2025 - 2/3

- If v_i is a sinusoidal voltage given by $4 \sin(t) V$, approximately sketch v_o vs. t when:
 - switch S_1 is closed and S_2 remains open
 - switch S_2 is closed and S_1 remains open



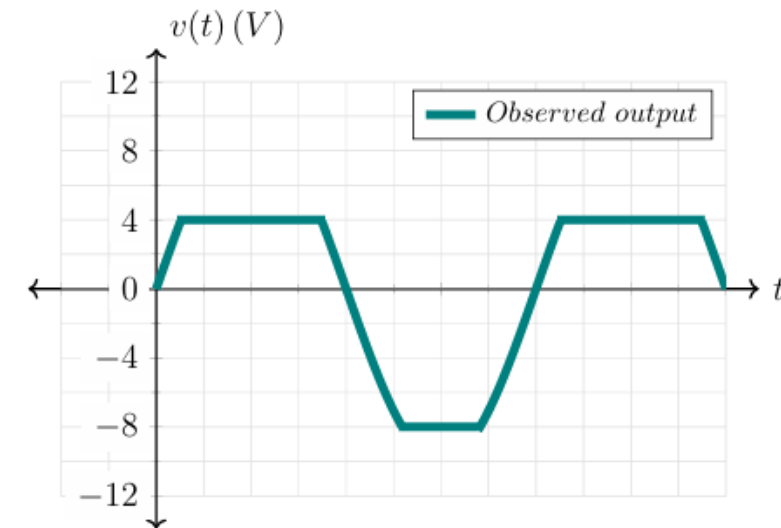
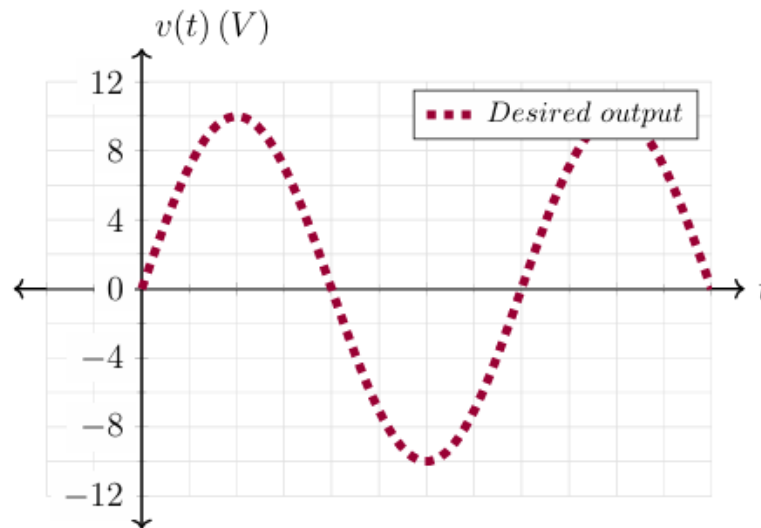
Mid — Summer 2025 - 3/3

- I. Implement the following function using op-amp circuits. Here, x , y , and z are analog voltage inputs, and f is the voltage output.

$$f = -5 \frac{d^2}{dt^2} x - \frac{5}{3} y + 7 \int z dt$$

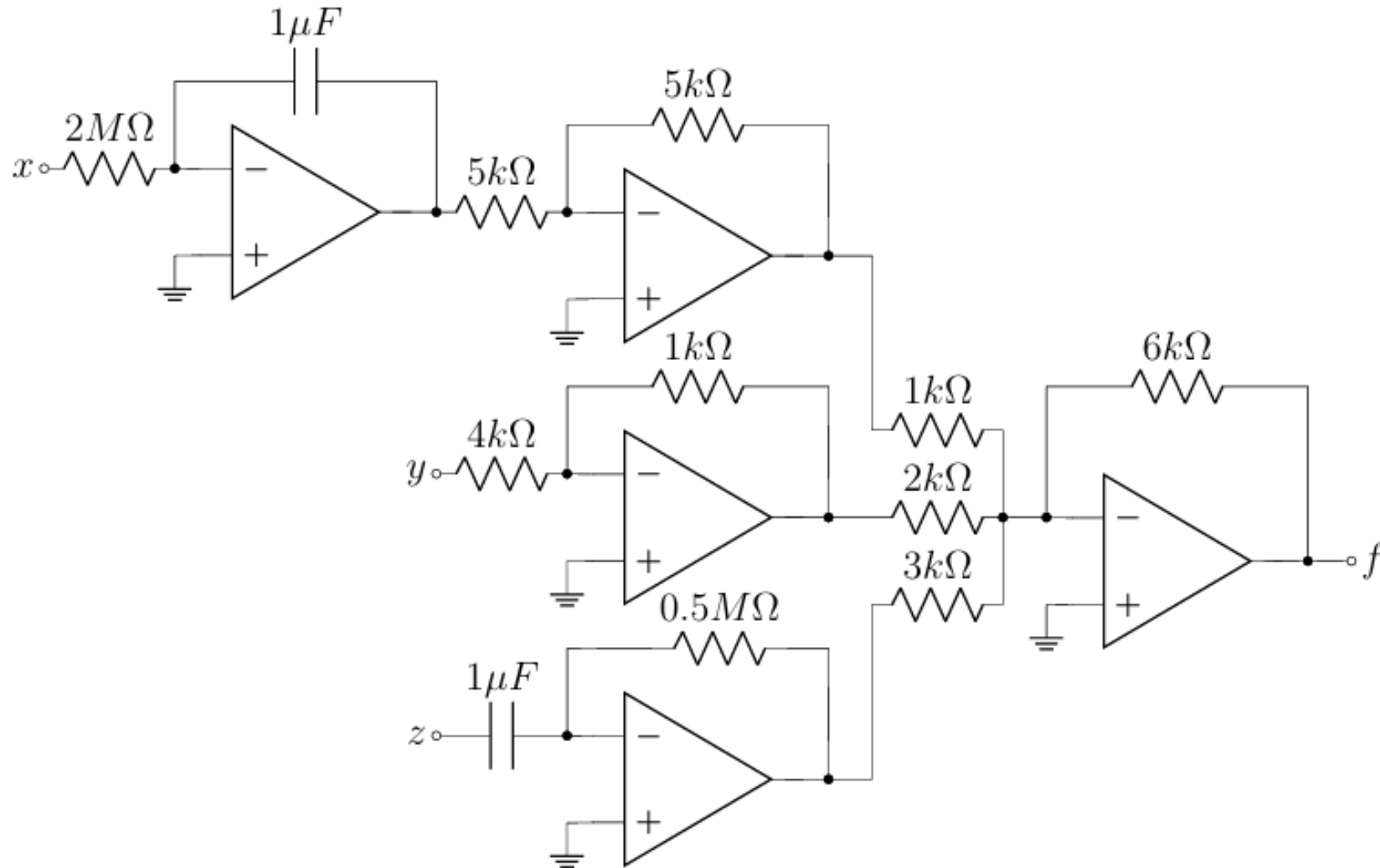
- II. An amplifier circuit was designed to amplify a weak sinusoidal voltage signal from a sensor, given by $2 \sin(t) V$, with a desired output waveform as shown in the left plot below. However, when the output voltage of the amplifier was observed on an oscilloscope, the waveform appeared as a distorted sinusoid, similar to the one shown on the right.

- a. Draw the amplifier circuit that produced the distorted sinusoidal output shown above, specifying all relevant component values.
- b. Sketch the VTC of the amplifier designed in part (a).



Mid — Spring 2025 - 1/2

- Express f as a function of variables x and y .

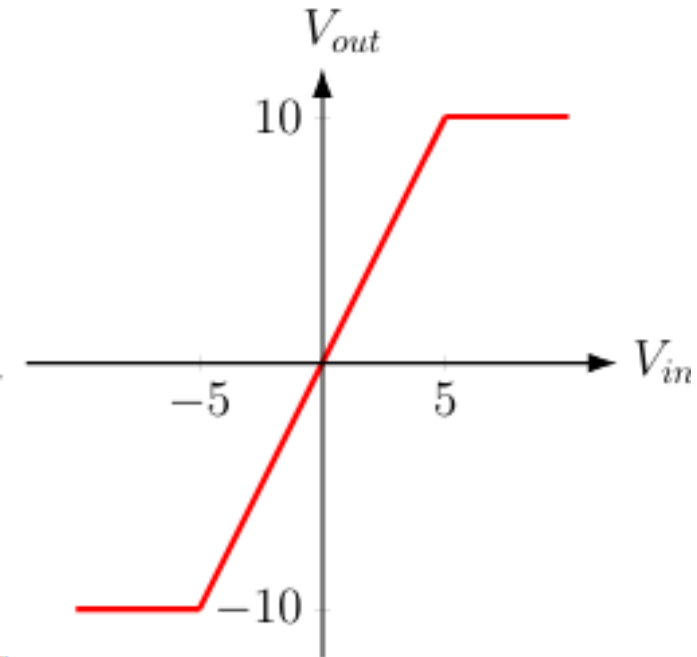
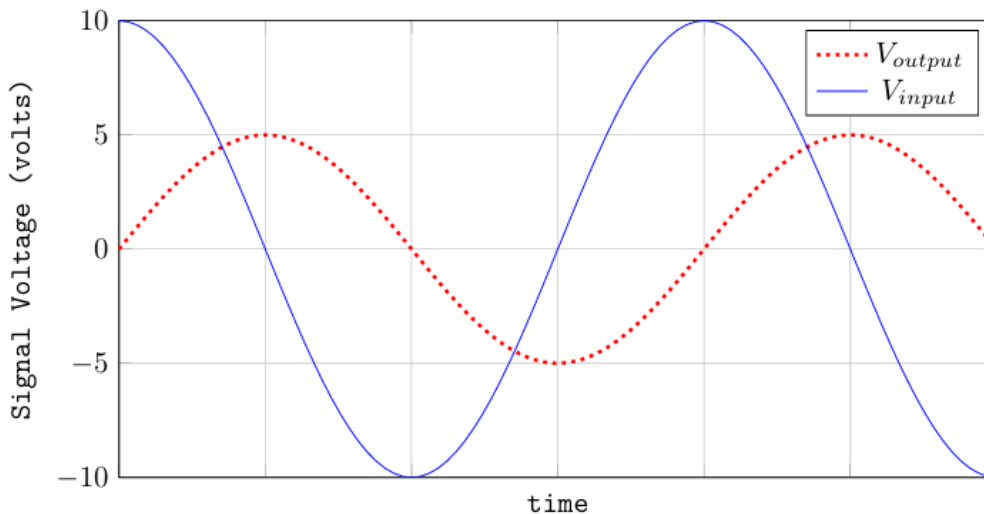


$$\text{Ans: } f = -3 \int x dt + \frac{3}{4} y - \frac{dz}{dt}$$

Mid — Spring 2025 - 2/2

- Analyze the plot on the left and design a circuit that implements the relationship between the voltage waveforms, V_{input} and V_{output} .
- Analyze the plot on the right and design a circuit that implements the relationship between V_{in} and V_{out} .
- Implement the following function using op-amp circuits. Here, x , y , and z are analog voltage inputs, and f is the voltage output.

$$f = 3 \frac{d}{dt}(x + y) - 5 \int z \, dt$$

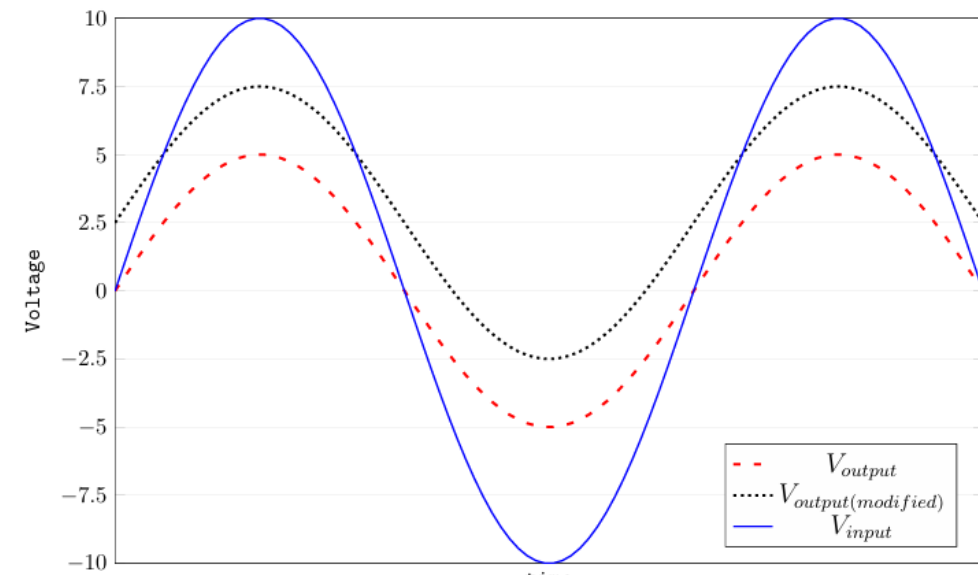
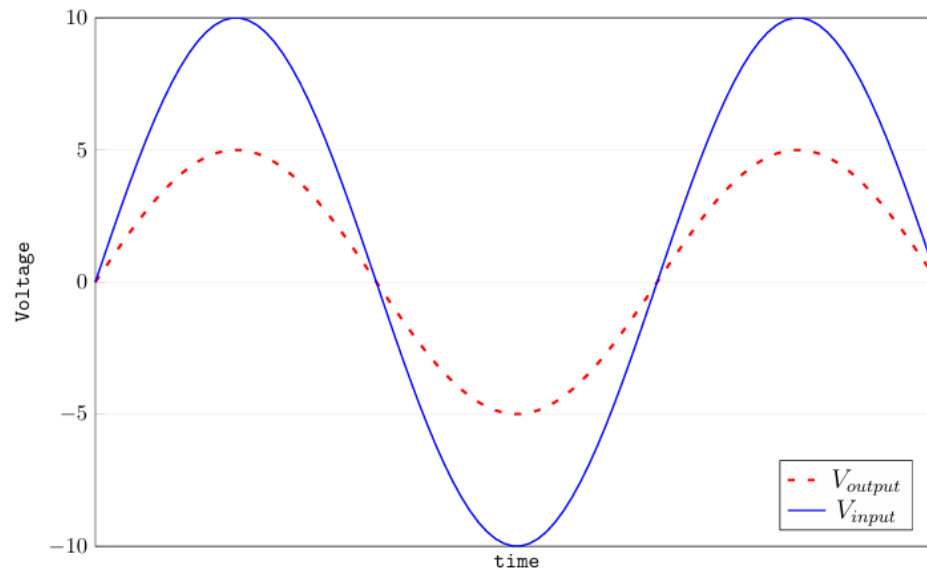


Mid — Fall 2024

- a. Implement the following function using op-amp circuits. Here, x , y , and z are analog voltage inputs, and f is the voltage output.

$$f = \frac{d^2}{dt^2}x + 10y + \int (10z - 9) dt$$

- b. Analyze the plot on the left and design a circuit that implements the relationship between the voltage waveforms, V_{input} and V_{output} .
- c. Modify the circuit you designed (b) so that $V_{output(modified)}$ represents the output waveform of the modified circuit.



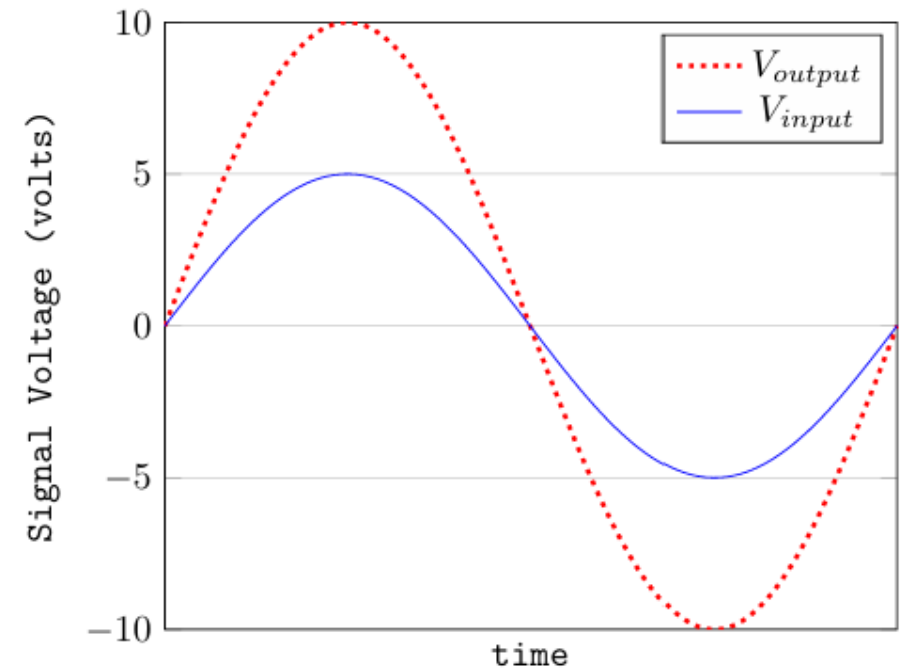
Mid — Summer 2024 - 1/2

- a. Michael Scott wants to make a phone call to his assistant, Dwight Schrute, but the outdated phone system keeps failing as the signal is too weak by the time it reaches Dwight. Dwight plans on designing a device to help Michael make the call. The device will take the weak signal as input and give a strong signal at its output without changing the waveshape and polarity of the signal as shown in the following figure.

- I. Draw the VTC of the device.
- II. Design a circuit for the device.

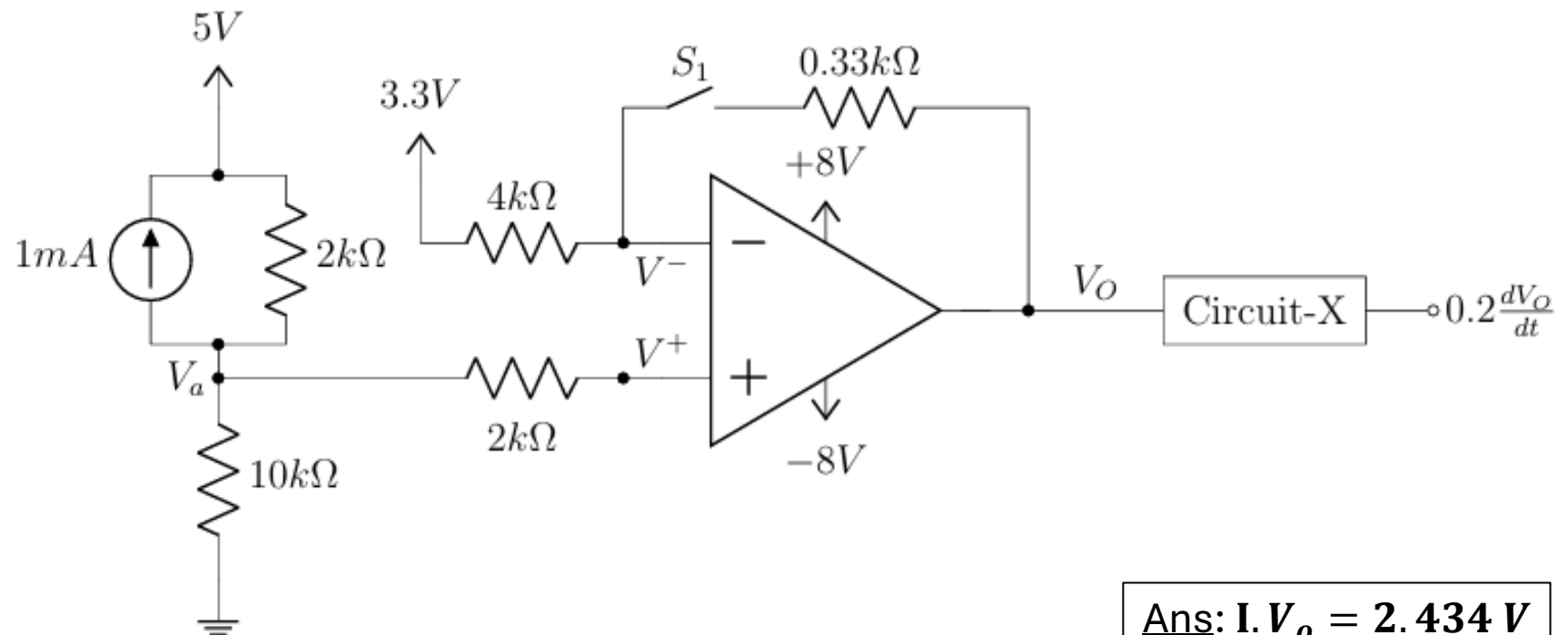
- b. Implement the following function using op-amp circuits. Here, x , y , and z are analog voltage inputs, and f is the voltage output.

$$f = -3 \frac{d}{dt} z + 6x + 9 \int y dt$$



Mid — Summer 2024 - 2/2

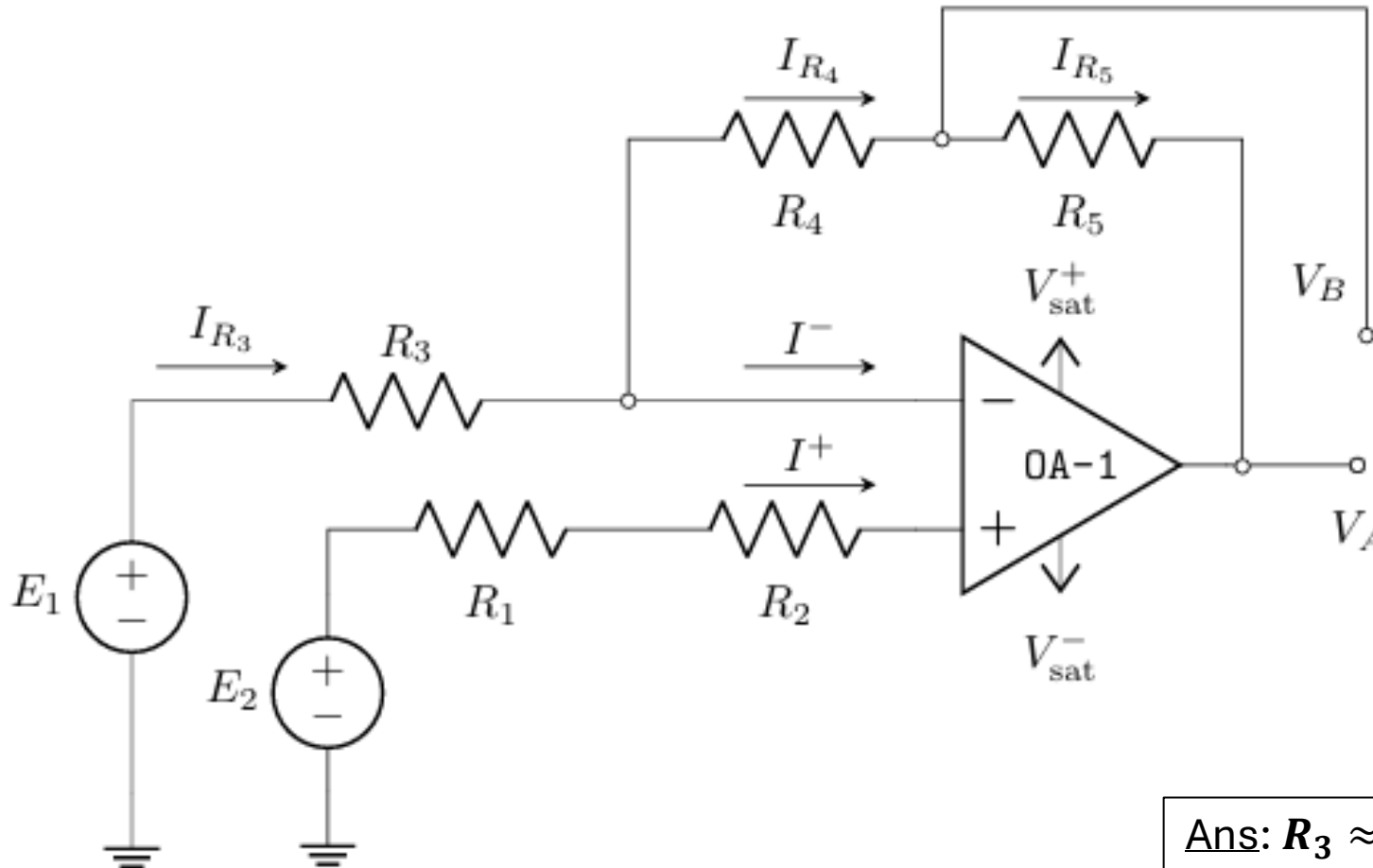
- The circuit diagram has a switch S_1 which is shown to be 'open' in the figure. The output V_O is passed through an unknown block of 'Circuit-X' and a differentiated result is generated.
 - Determine V_O when the switch S_1 is closed.
 - Design the Circuit-X.



Ans: I. $V_O = 2.434 V$

Mid — Spring 2024

- For $E_1 = 0\text{ V}$ and $E_2 = 2.2\text{ V}$ in the following circuit, V_A was measured to be 5.13 V . Determine the value of R_3 and V_B .

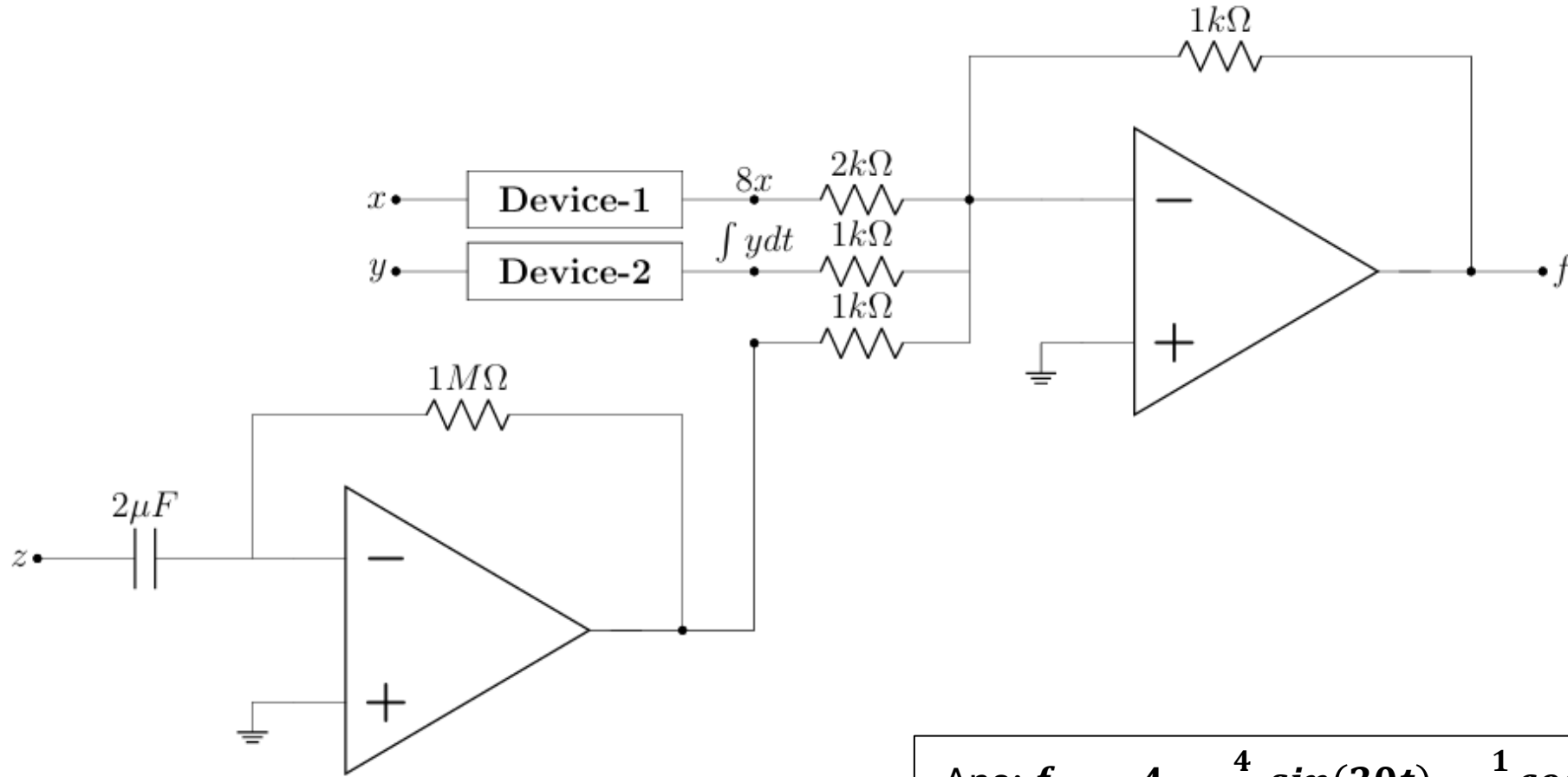


$$\begin{aligned} V_{Sat}^+ &= +8\text{ V} \\ V_{Sat}^- &= -8\text{ V} \\ R_1 &= 1\text{ k}\Omega \\ R_2 &= 1\text{ k}\Omega \\ R_4 &= 5\text{ k}\Omega \\ R_5 &= 2\text{ k}\Omega \end{aligned}$$

Ans: $R_3 \approx 5.25\text{ k}\Omega$, $V_B \approx 2.5\text{ V}$

Mid — Summer 2023

- If $x = 1\text{ V}$, $y = 8 \cos(30t)\text{ V}$, and $z = \sin(4t)\text{ V}$, determine f as a function of t .



$$\text{Ans: } f = -4 - \frac{4}{15} \sin(30t) - \frac{1}{2} \cos(4t) \text{ V}$$

Mid — Spring 2023 - 1/2

- a. Farhan, CEO of Rajshahi WASA, is building a water level indicator for an overhead tank. For this, he placed three sensors at three different levels of the tank. The voltage outputs of the three sensors are denoted as V_1 (lowest level), V_2 (mid-level), and V_3 (highest level). Farhan decided that the indicator for the water level would be,

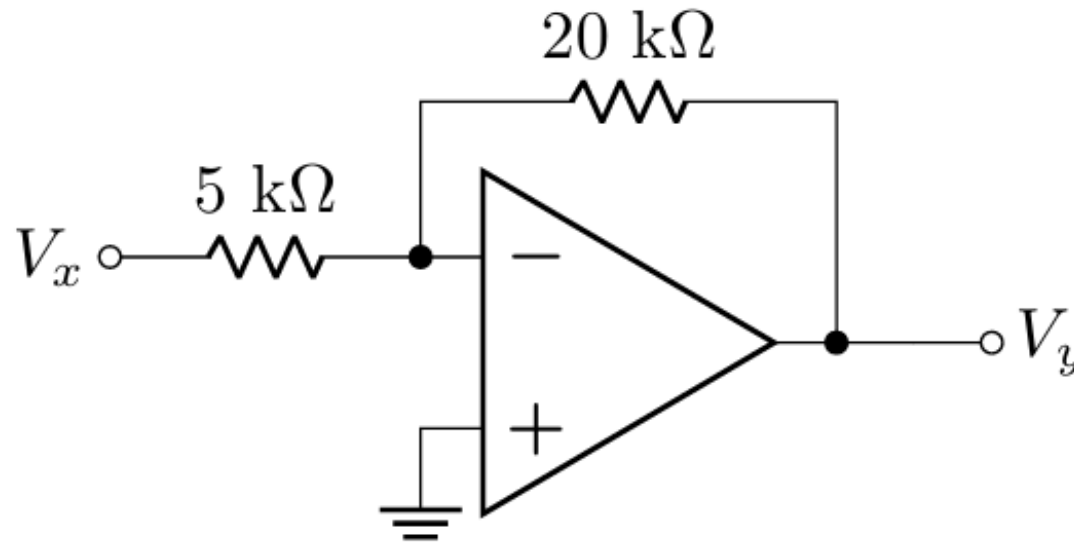
$$V_{indicator} = \frac{1}{k} (V_1 + V_2 + V_3)$$

Assuming $k = 1$, design the circuit using op amps that will take V_1 , V_2 , and V_3 as input and will produce $V_{indicator}$ as output.

- b. Upon further experimentation, Farhan realized that the maximum values of V_1 , V_2 , and V_3 are 5 V . Hence, the maximum value of $V_1 + V_2 + V_3$ can be 15 V . However, Farhan only has access to $+5\text{ V}$ and -5 V as power supplies, meaning that the $V_{indicator}$ cannot be greater than $+5\text{ V}$ or less than -5 V .
- I. Choose a value of k such that the maximum value of $V_{indicator}$ is within the given range.
 - II. Design the circuit in part (a) again using this new value of k .

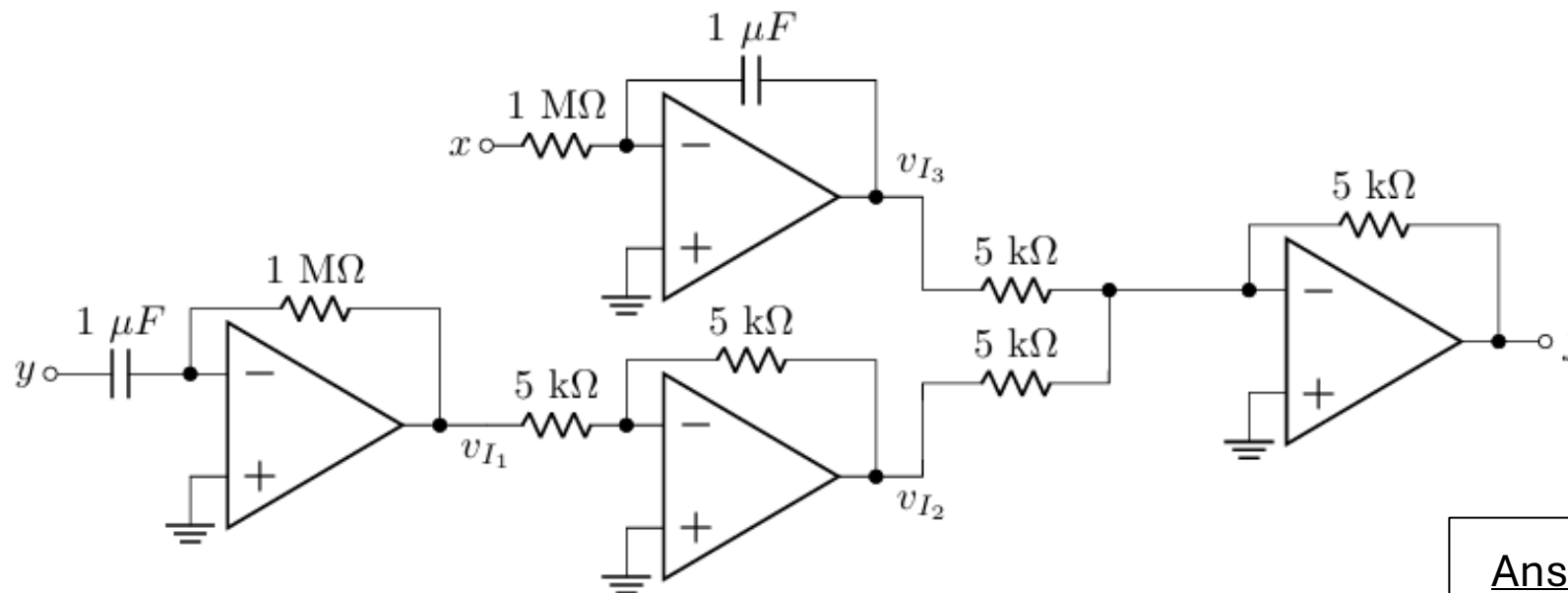
Mid — Spring 2023 - 2/2

- Design a circuit using op amp that takes $V_i = 1 - 0.5 \cos t$ as input and produces $V_o = 2 \sin t$ as output.
- Now, the output V_o of part (a) is used as the input to the circuit shown in the following figure, i.e., $V_x = V_o = 2 \sin t$. Approximately sketch V_y as a function of t .



Final — Fall 2022

- Derive an expression for the output voltage f in terms of the input voltages x and y .
- Draw the circuit of an inverting amplifier and design it in such a way that the voltage gain is $k = -4$.
- Assuming a sinusoidal input of 0.5 V amplitude, approximately sketch the input and output waveforms of the inverting amplifier designed in part (b).



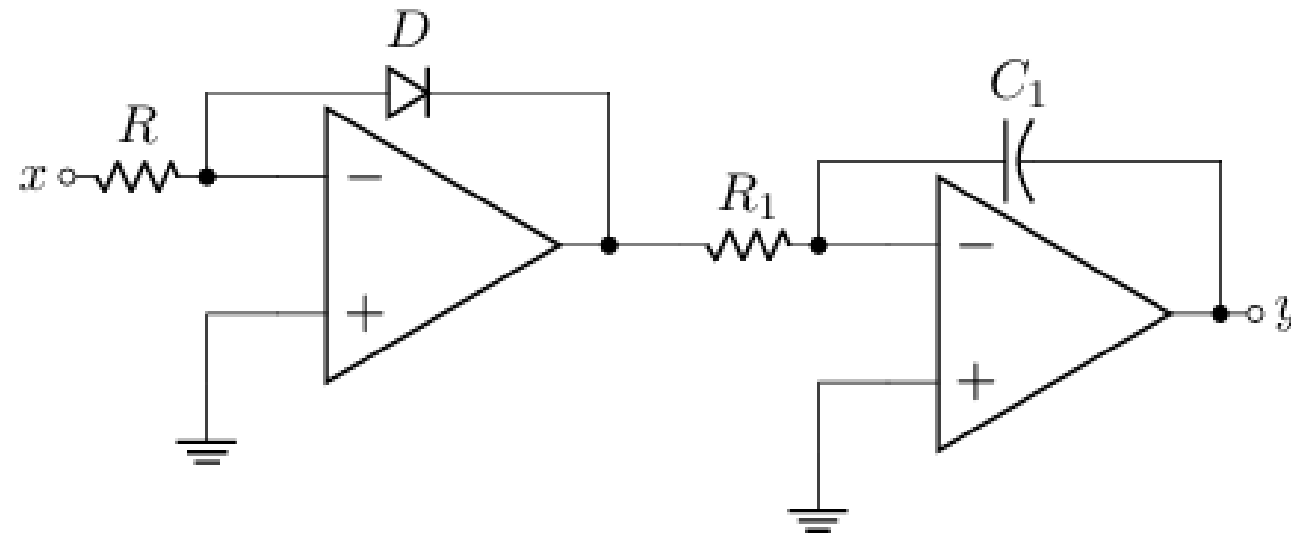
$$\text{Ans: (a) } f = \int x \, dt - \frac{dy}{dx}$$

Final — Summer 2022

- a. Implement the following function using op-amp circuits. Here, x , y , and z are analog voltage inputs, and f is the voltage output.

$$f = -3 \frac{d}{dt} x + 2e^y + 4z$$

- b. In the following circuit, find y in terms of x . For simplicity, assume $V_T = 1$ and $I_S R = 1$, where V_T is the thermal voltage and I_S is the reverse saturation current for the diode D .



Acknowledgement and References

Some of the problems in this set are taken or adapted from the following sources:

1. Sedra, A. S., & Smith, K. C., Microelectronic Circuits, Oxford University Press
2. Coughlin, R. F., & Driscoll, F. F., Operational Amplifiers and Linear Integrated Circuits, Pearson
3. Neamen, D. A., Microelectronics: Circuit Analysis and Design, McGraw-Hill